



Review article

Targeting the mitochondrial functional network: Decoding upstream pathological mechanisms and novel therapeutic paradigms for Alzheimer's disease

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ABSTRACT

Alzheimer's disease (AD) is the most prevalent neurodegenerative disorder worldwide. Conventional downstream interventions targeting β -amyloid (A β) and tau proteins have repeatedly failed in clinical practice, and mitochondrial functional decline has been identified as the core upstream driver of AD pathogenesis. Focusing on the mitochondrial functional network as the core target, this paper systematically dissects the key molecular mechanisms of mitochondrial dysfunction during AD progression, including oxidative phosphorylation impairment, mitochondrial DNA (mtDNA) mutations and genetic defects, excessive reactive oxygen species (ROS) production, mitochondrial dynamics imbalance, mitophagy dysfunction, calcium homeostasis dysregulation, mitochondrial transport defects, and the bidirectional regulatory pathway of tau pathology, and elucidates the pathological network featured by cascade amplification and reciprocal regulation among these abnormal mechanisms. It also comprehensively summarizes mitochondria-targeted intervention strategies for AD, analyzes the research limitations in this field such as model heterogeneity, lack of specific biomarkers and inefficient drug delivery, and prospects future research directions by integrating cutting-edge technologies including cell reprogramming and artificial intelligence (AI). This study provides a novel interpretation of the aging-related pathogenic mechanisms of AD from a mitochondrial perspective, lays a theoretical foundation for the development of precise and efficient mitochondria-targeted therapeutic strategies for AD, and offers a new paradigm for breaking through the bottlenecks of clinical diagnosis and treatment of AD.

1. Introduction

Alzheimer's disease (AD), the most prevalent neurodegenerative disorder, exhibits rising incidence with global population aging and constitutes a major public-health burden (Burns et al., 2022). The growing number of AD patients imposes substantial economic and care-giving pressure on families and healthcare systems worldwide (Wimo et al., 2023). Despite decades of extensive research on the classic pathological hallmarks of AD such as β -amyloid (A β) deposition and hyperphosphorylation of tau protein (Reddy and Oliver, 2019), most clinical trials targeting these downstream pathological processes have ended in failure. This indicates that the pathogenesis of AD is far more complex than previously understood, and there is an urgent need to

identify core driving factors from the perspective of upstream regulatory mechanisms.

As the energy organelle of eukaryotic cells, mitochondria generate adenosine triphosphate (ATP) via oxidative phosphorylation and participate in oxidative stress, calcium homeostasis, apoptosis and signal transduction (Wang et al., 2024b). The functional integrity of mitochondria is of particular importance in nerve cells. As highly energy-demanding cells, neurons rely heavily on mitochondrial energy supply for physiological activities such as synaptic transmission and axonal transport (Harris et al., 2012). The limited regenerative capacity and antioxidant reserves in the central nervous system further amplify the damaging effects of mitochondrial dysfunction on neurons. In recent years, a growing body of research has confirmed a close temporal

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correlation between mitochondrial senescence and the pathological progression of AD (Oliver and Reddy, 2019a; Reddy, 2009b). Studies have shown that early signs of mitochondrial dysfunction can be detected in the brains of patients decades before the onset of clinical AD symptoms. With the gradual emergence of core senescent features such as reduced mitochondrial biogenesis capacity, dysfunction of the quality control system and impairment of functional homeostasis, classic pathological changes including A β deposition and abnormal tau aggregation subsequently occur, ultimately leading to neuronal loss and cognitive impairment (Bigio et al., 2025; Cantero et al., 2025; D'Alessandro et al., 2025).

Extensive basic and clinical studies have provided solid evidence for the core hypothesis that mitochondrial dysfunction acts as an upstream driver of AD. Substantial pre-clinical and clinical evidence supports mitochondrial dysfunction as an upstream driver of AD. Post-mortem AD brain tissues show abnormal mitochondrial ultrastructure in cortex and hippocampus, including disrupted cristae, swelling and rupture, accompanied by reduced oxidative phosphorylation (OXPHOS) complex activity and elevated mitochondrial DNA mutation load (Baloyannis, 2006; Gu et al., 2024). Animal experiments have confirmed that inducing mitochondrial dysfunction in mouse neurons through gene editing techniques directly leads to increased A β production, abnormal tau phosphorylation and cognitive decline. In contrast, ameliorating mitochondrial function can effectively slow down the pathological progression and behavioral deficits in AD model animals (Du et al., 2017; Kukreja et al., 2014). Clinical studies have also demonstrated the presence of mitochondrial dysfunction in peripheral blood leukocytes and skin fibroblasts of AD patients, and the severity of such dysfunction is positively correlated with the degree of cognitive impairment (Bosetti et al., 2002; Fišar et al., 2016). Collectively, these lines of evidence suggest that mitochondrial senescence may represent a core upstream

node in AD pathogenesis, and the multidimensional dysfunction mediated by mitochondrial senescence constitutes a key driving force for the pathological progression of AD (Fig. 1).

This review focuses on how mitochondrial senescence modulates AD pathogenesis. We systematically summarize mitochondrial-related mechanisms and their crosstalk networks, as well as mitochondria-targeted therapeutic strategies for AD. This work aims to provide new insights into AD pathogenesis and theoretical references for developing precise preventive and therapeutic approaches for AD.

2. Mitochondrial regulatory core mechanisms in AD pathogenesis

In AD, mitochondrial dysfunction occurs throughout disease progression, covering multiple key pathological events: OXPHOS defects, mtDNA mutations and genetic abnormalities, over-production of ROS, disturbed mitochondrial dynamics, impaired mitophagy, disrupted calcium homeostasis, and defective mitochondrial trafficking. These pathological alterations mutually regulate each other to create a vicious cycle, and crosstalk with canonical A β and tau pathologies to collectively trigger neuronal injury, synaptic degeneration and cognitive impairment. Unraveling these mechanisms is critical for AD diagnosis and therapeutic development (Fig. 2).

2.1. Mitochondrial oxidative phosphorylation impairment

OXPHOS represents the core cellular energy-metabolic process occurring at the inner mitochondrial membrane: the electron transport chain establishes a proton gradient to drive ATP production (Nunnari and Suomalainen, 2012; Rossmann et al., 2021). In neurons, over 95% of cellular ATP derives from OXPHOS. At synapses, mitochondria provide

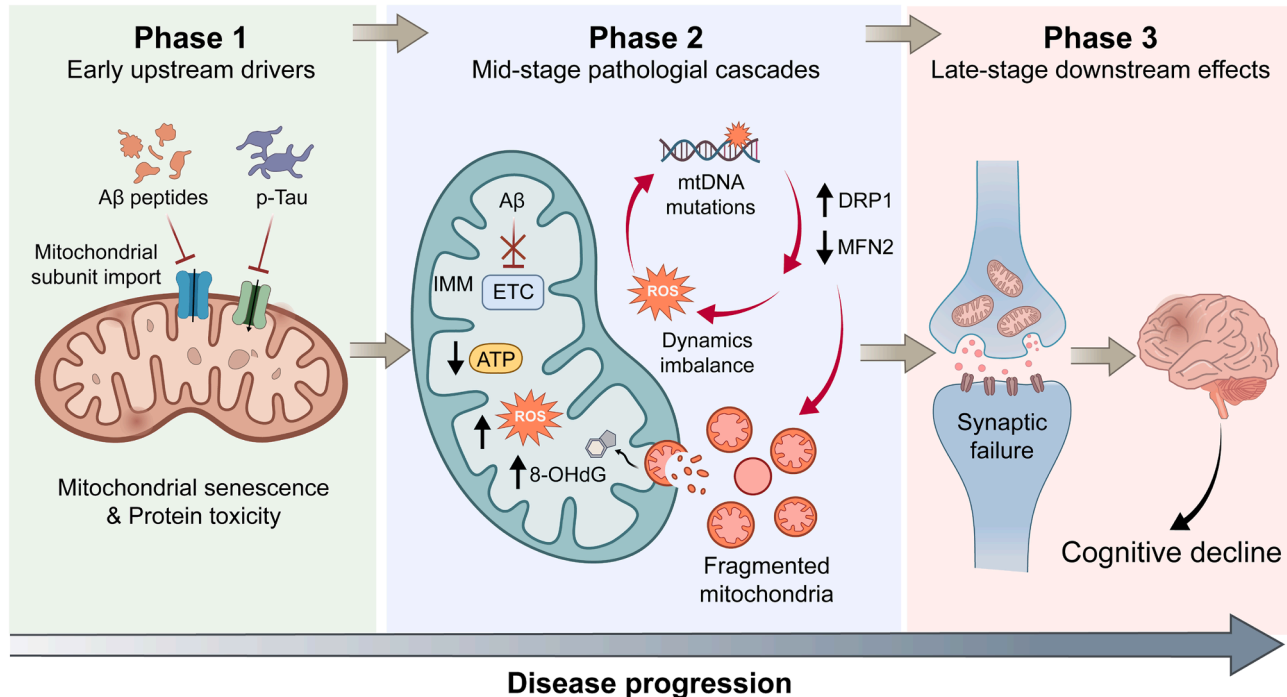


Fig. 1. Chronological progression of mitochondrial dysfunction-driven pathological cascades in AD. In the early stage, abnormally accumulated A β and p-Tau block the import of mitochondrial subunits and trigger mitochondrial stress and senescence. In the middle stage, A β inhibits the electron transport chain, impairing ATP production and inducing sharp surges of ROS and 8-OHdG. Meanwhile, mtDNA mutations, oxidative stress, and dysregulated mitochondrial dynamics (elevated DRP1 / reduced MFN2) form a vicious amplifying cascade that disrupts the mitochondrial network into dysfunctional fragmented mitochondria. In the late stage, these fragmented mitochondria abnormally accumulate within synapses, causing synaptic failure and ultimately irreversible cognitive decline. AD (Alzheimer's disease), A β (β -amyloid), p-Tau (phosphorylated tau), IMM (inner mitochondrial membrane), ETC (electron transport chain), ATP (adenosine triphosphate), ROS (reactive oxygen species), 8-OHdG (8-hydroxy-2'-deoxyguanosine), mtDNA (mitochondrial DNA), DRP1 (dynamin-related protein 1), MFN2 (mitofusin 2). Figure created with Adobe Illustrator.

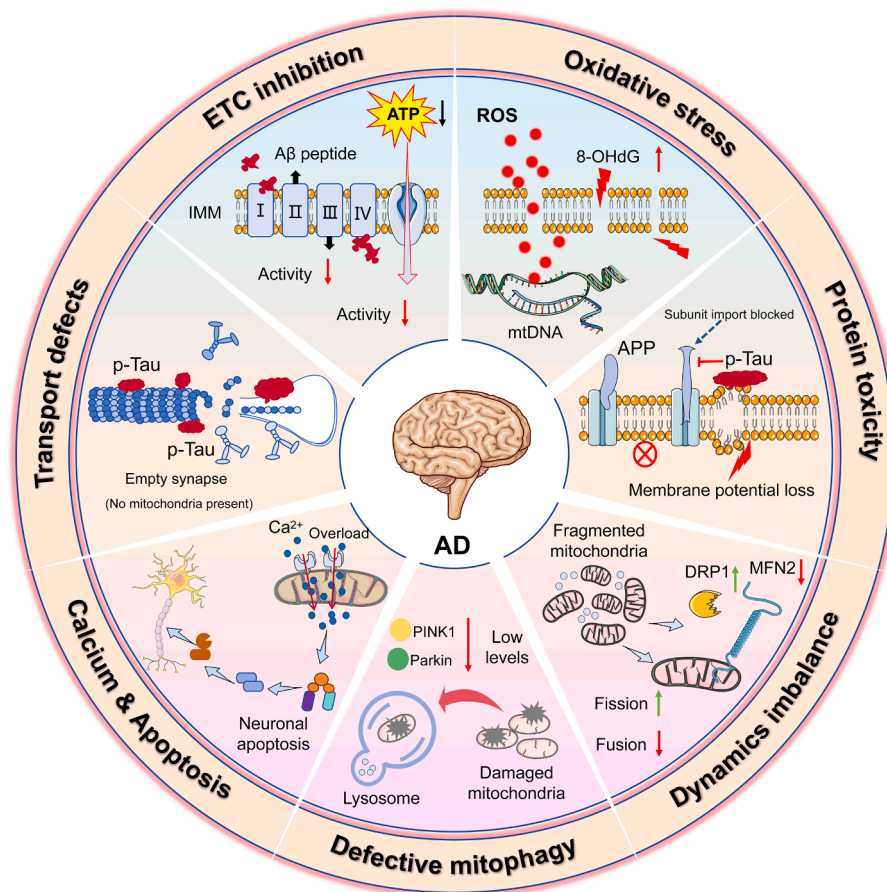


Fig. 2. Core mechanisms of mitochondrial dysfunction-driven pathological cascades in AD. In AD, the pathological accumulation of A β peptide and p-Tau drives a network of mitochondrial impairments that culminate in synaptic failure and cognitive decline. A β peptide inhibits the ETC at the IMM, reducing ATP production and mitochondrial activity. Concurrently, elevated ROS and 8-OHdG induce oxidative stress, damaging mtDNA and cellular structures. Protein toxicity arises as A β and p-Tau disrupt APP processing, block mitochondrial subunit import, and cause membrane potential loss. Mitochondrial dynamics are imbalanced, with increased fission (mediated by DRP1) and decreased fusion (mediated by MFN2), leading to fragmented mitochondria. Defective mitophagy, driven by reduced PINK1 and Parkin levels, impairs the clearance of damaged mitochondria, while Ca²⁺ overload triggers neuronal apoptosis. Additionally, p-Tau accumulation disrupts mitochondrial transport, resulting in empty synapses devoid of functional mitochondria. Together, these interconnected pathological processes disrupt synaptic integrity and neuronal function, contributing to the progression of AD. AD (Alzheimer's disease), A β (β -amyloid), p-Tau (phosphorylated tau), ETC (electron transport chain), IMM (inner mitochondrial membrane), ROS (reactive oxygen species), 8-OHdG (8-hydroxy-2'-deoxyguanosine), Ca²⁺ (calcium), mtDNA (mitochondrial DNA), APP (amyloid precursor protein), DRP1 (dynamins-related protein 1), Mfn2 (mitofusin 2), PINK1 (PTEN-induced kinase 1), Parkin (Parkinson disease protein 2). Figure created with PowerPoint.

rapid ATP supply to sustain high-energy processes including presynaptic neurotransmitter release, postsynaptic ion-channel activity and synaptic plasticity (Attwell and Laughlin, 2001; Harris et al., 2012; Li et al., 2026b).

In the pathological progression of AD, abnormal mitochondrial ETC function constitutes the core link of OXPHOS impairment and has been identified as an early pivotal event in disease onset (Mao and Reddy, 2011). Among them, electron leakage from Complexes I and III is the major source of ROS production (Reddy, 2008). Multiple studies have confirmed that the activity of mitochondrial OXPHOS complexes is significantly reduced in cognition-related brain regions of AD patients, such as the cerebral cortex, hippocampus and amygdala, with the damage to Complexes I and IV being the most prominent (Lin and Beal, 2006; Reddy et al., 2010). A study has found a marked decrease in the activity of mitochondrial Complexes I and IV in the frontal cortex of AD patients, and this reduction is not caused by a decline in mitochondrial quantity but by intrinsic structural and functional abnormalities of the complexes themselves (Kenney and Bennett, 2019). Subsequent research further verified that the activity of mitochondrial Complexes I and IV is impaired to varying degrees in the hippocampus of AD patients, with the most significant decline in Complex IV activity observed in the

hippocampal CA1 region (Parker et al., 1994). Notably, OXPHOS dysfunction exhibits distinct regional specificity in the brain: mitochondrial damage in cognition-related brain regions of AD patients, such as the cerebral cortex and hippocampus, is far more severe than that in other brain regions, and the degree of such damage is positively correlated with the density of neuronal loss, A β deposition and tau tangles (Simonian and Hyman, 1995). This region-specific damage is closely associated with the direct action of core AD pathological factors: A β peptides can directly penetrate the double mitochondrial membrane and bind to the subunits of Complex IV, significantly inhibiting its catalytic activity. This leads to a notable reduction in ATP production efficiency and a concomitant explosive accumulation of ROS (Reddy, 2009a). Further observations via immunoelectron microscopy have revealed the colocalization of A β and amyloid precursor protein (APP) in hippocampal neuronal mitochondria of AD patients. APP can block the translocation of nuclear-encoded cytochrome c oxidase subunits to mitochondria, thereby forming a vicious cycle of ETC inhibition and elevated ROS levels. This mechanism is commonly expressed in both early-onset familial AD (FAD) and sporadic AD (SAD) (Devi et al., 2006). In addition to A β , hyperphosphorylated tau protein is another important pathological factor that exacerbates OXPHOS dysfunction. It can

interact with mitochondrial membrane proteins to disrupt the integrity of the inner mitochondrial membrane and reduce mitochondrial membrane potential, thus further inhibiting the normal exertion of OXPHOS function (Torres et al., 2022).

Reduced activity of OXPHOS complexes directly leads to the failure of neuronal energy metabolism. ATP levels are significantly decreased in the brains of AD patients, and insufficient ATP supply at synaptic sites directly impairs the release of presynaptic neurotransmitter vesicles, the function of postsynaptic ion pumps, as well as the synthesis and modification of synaptic plasticity-related proteins (Walker, 2025). This consequently results in reduced synaptic transmission efficiency, synaptic dysfunction and even synaptic loss. Studies have shown that mitochondrial ATP production is markedly diminished in synaptosomes of AD model mice, and supplementation with exogenous ATP can partially restore synaptic transmission function and ameliorate cognitive impairment in these mice (Wang et al., 2016). Furthermore, OXPHOS dysfunction causes ETC blockage, which increases electron leakage and thus promotes ROS generation, further exacerbating neuronal damage (Turrens, 2003). The occurrence of OXPHOS dysfunction in AD is not mediated by a single factor; its molecular mechanisms are highly complex and involve multiple levels of pathological changes, including abnormal expression of complex subunits, dysregulated post-translational modifications, coenzyme deficiency and decreased mitochondrial membrane potential. In the mitochondria of AD patients and model animals, the expression levels of genes encoding OXPHOS complex subunits, including both nuclear and mtDNA-encoded genes, are significantly downregulated (Abolhassani et al., 2017; Manczak et al., 2004). Meanwhile, AD-related pathological factors such as A β and hyperphosphorylated tau protein can directly bind to OXPHOS complexes and disrupt their structural integrity and functional stability (Devi et al., 2006; Jia et al., 2025). A decreased mitochondrial membrane potential is an important marker and consequence of OXPHOS dysfunction: the mitochondrial membrane potential is significantly reduced in the brains of AD patients, leading to insufficient formation of the transmembrane proton gradient and inhibited ATP synthase activity, which further aggravates the failure of energy metabolism (Seelert et al., 2009). Clinical studies have also provided strong evidence for the association between OXPHOS dysfunction and AD. Multiple studies have found that the activity of mitochondrial OXPHOS complexes is similarly reduced in peripheral blood leukocytes, skin fibroblasts and platelets of AD patients, and the degree of this reduction is negatively correlated with patients' cognitive function scores, suggesting that abnormal peripheral mitochondrial OXPHOS function may serve as a potential biomarker for AD (Khan and Alkon, 2015; Mahapatra et al., 2023; Parker et al., 1990). In addition, individuals carrying AD susceptibility genes such as the APOE ϵ 4 allele already exhibit mild abnormalities in mitochondrial OXPHOS function in both brain and peripheral cells, indicating that OXPHOS dysfunction may be an early event in AD pathogenesis, preceding the emergence of classic pathological hallmarks (Jairani et al., 2019).

In summary, mitochondrial OXPHOS dysfunction underlies cerebral energy metabolic failure in AD, driving AD progression via ATP deficiency, excessive ROS production and aggravated synaptic damage. Deciphering the molecular mechanisms of OXPHOS dysfunction and its crosstalk with AD pathological hallmarks is critical for revealing AD pathogenesis and developing targeted therapeutic strategies.

2.2. Mitochondrial mtDNA mutations and genetic defects

mtDNA is a circular double-stranded extra-nuclear genome (~16.5 kb) that encodes key components for OXPHOS complexes and mitochondrial translation machinery (Anderson et al., 1981; Taanman, 1999). Unlike nuclear DNA, mtDNA is not protected by histones and resides in a high-ROS microenvironment; its limited repair capacity confers a 10- to 20-fold higher mutation rate (Brown et al., 1979). mtDNA mutations can be maternally inherited or somatically

accumulated over lifetime under oxidative stress, raising mtDNA mutation load with advancing age (Giles et al., 1980; Lee et al., 2023).

Extensive studies have confirmed a marked elevation in mtDNA mutation load in the brain and peripheral tissues of AD patients, accompanied by the presence of various specific mtDNA mutation types (Mao et al., 2012; Phillips et al., 2014). Early sequencing analyses of mtDNA from the brains of AD patients identified a large number of point mutations, deletion mutations and insertion mutations in the mtDNA coding region. Among these, mutations in genes encoding subunits of Complex I, Complex IV and tRNA genes are the most prevalent (Keogh and Chinnery, 2015). Studies have demonstrated that mutations in the ND4 gene in the brains of AD patients impair Complex I activity and blocks electron transport. Mutations in the COX1 gene disrupt the structure and function of Complex IV, leading to reduced OXPHOS efficiency (Hou et al., 2017; Lee et al., 2018; Monzio Compagnoni et al., 2020). Furthermore, various deletion mutations exist in the mtDNA of AD patients, such as the 4977 bp deletion and 7436 bp deletion. These deletion mutations result in the loss of multiple genes encoding OXPHOS subunits, directly impairing the assembly and function of OXPHOS complexes (Zheng et al., 2012). These AD-enriched mtDNA mutations are not mere by-products of AD pathogenesis; rather, they directly trigger OXPHOS defects and ROS overproduction via multiple molecular mechanisms and act as key intrinsic drivers of AD progression. First, mtDNA-encoded subunits constitute core components of OXPHOS complexes. Mutations alter subunit structure, compromise complex assembly and enzymatic activity, suppress electron transport chain function, lower OXPHOS efficiency and curtail ATP synthesis (Wallace, 2010). For instance, mutations in the ND1 gene lead to a significant reduction in Complex I activity, blocking electron transport and causing a marked decline in ATP synthesis (Mao et al., 2008). For instance, mutations in the ND1 gene lead to a significant reduction in Complex I activity, reducing electron transport efficiency and resulting in OXPHOS dysfunction (Hock et al., 2020). Secondly, electron transport chain blockage caused by mtDNA mutations increases electron leakage, thereby promoting ROS generation. Under normal physiological conditions, approximately 1–2% of electrons in the electron transport chain leak and combine with oxygen molecules to form superoxide anions (O $_2^{\cdot -}$). During OXPHOS dysfunction, the electron leakage rate increases significantly, leading to massive ROS production (Turrens, 2003). Excessive ROS not only further damages mtDNA, forming a vicious cycle of mtDNA mutation, ROS generation and further mtDNA damage, but also causes oxidative damage to mitochondrial proteins, lipids and nuclear DNA, disrupting intracellular homeostasis and ultimately leading to neuronal injury (Nunomura et al., 2001).

Moreover, mtDNA mutations aggravate OXPHOS defects through disrupting mitochondrial protein synthesis. mtDNA-encoded tRNAs and rRNAs are indispensable for this process; tRNA-gene mutations induce tRNA structural defects, compromise amino acid delivery and codon recognition, repress the synthesis of mtDNA-encoded OXPHOS subunits, and ultimately result in defective complex assembly (Taanman, 1999). A study has identified the mtDNA tRNA gene mutation in AD patients, which results in structural abnormalities of the amino acid acceptor stem of tRNA, impairs its binding efficiency to leucine, inhibits mitochondrial protein synthesis and thus reduces the expression levels of OXPHOS complexes (Silzer et al., 2020; Wallace, 1994). Meanwhile, mtDNA mutations trigger mitochondrial stress responses and activate the regulatory expression of nuclear genes, such as the modulation of mitochondrial biogenesis via the PGC-1 α pathway. However, long-term mtDNA damage leads to an imbalance in stress responses, further exacerbating mitochondrial dysfunction (Halling and Pilegaard, 2020). Genetic epidemiological studies also support the association between mtDNA mutations and AD. Studies have found that certain mtDNA haplogroups are significantly associated with the risk of AD onset. For example, mtDNA haplogroups H, J and K are associated with a reduced risk of AD, while haplogroups U and T may increase this risk (Maruszak et al., 2011; Valentino et al., 2022). In addition, individuals carrying

mtDNA mutations may develop AD at an earlier age and experience a faster disease progression (Bi et al., 2015). These studies suggest that mtDNA mutations may be an important genetic risk factor for AD, influencing disease susceptibility and progression by regulating mitochondrial function. Clinical studies have also found that mtDNA mutation load is significantly higher in the peripheral blood, cerebrospinal fluid (CSF) and CSF cells of AD patients than in healthy controls, and the mutation load is positively correlated with the degree of cognitive impairment and disease duration in AD patient (Risi et al., 2025). For example, the frequency of the 4977 bp deletion mutation in mtDNA is markedly elevated in the CSF of AD patients and increases progressively with disease progression, indicating that mtDNA mutations may serve as a potential biomarker for monitoring AD progression (Casoli et al., 2015). Furthermore, studies on the family members of AD patients have shown that carriers of mtDNA mutations, although not yet presenting with clinical AD symptoms, already exhibit mild cognitive decline and mitochondrial dysfunction, suggesting that mtDNA mutations may be an early warning sign of AD onset (Onyango et al., 2021).

In conclusion, mtDNA mutations and genetic defects represent critical upstream drivers of OXPHOS impairment and ROS overproduction. They contribute to AD pathogenesis via disrupting mitochondrial energy metabolism, inducing oxidative stress and aggravating neuronal injury. Characterizing the types, distribution profiles and pathogenic mechanisms of mtDNA mutations is of theoretical and clinical importance for uncovering AD genetic predisposition, identifying early-diagnostic biomarkers and designing targeted therapeutic strategies.

2.3. Excessive mitochondrial ROS production

ROS refer to a class of highly reactive oxygen derivatives generated during cellular metabolism, mainly including O_2^- , hydrogen peroxide (H_2O_2), and hydroxyl radicals ($\bullet OH$) (Turrens, 2003). Under physiological conditions, ROS production and scavenging are dynamically balanced; moderate ROS serves as signaling molecules regulating cell proliferation, differentiation and apoptosis. Excessive ROS or impaired antioxidant clearance induces oxidative damage to cellular macromolecules, provokes oxidative stress and disturbs cellular homeostasis. Such oxidative stress occupies a central position in AD pathogenesis (Dewanjee et al., 2022; Reddy, 2009b).

Robust oxidative stress damage occurs in AD brains and precedes canonical A β deposition and tau neurofibrillary tangles. Post-mortem studies demonstrate increased oxidative-stress markers in AD-vulnerable brain regions including the cerebral cortex and hippocampus. Levels of lipid peroxidation products (e.g., malondialdehyde, 4-hydroxynonenal), protein oxidation products (e.g., 3-nitrotyrosine, carbonylated proteins), and DNA oxidation products (e.g., 8-hydroxy-2'-deoxyguanosine) are markedly elevated relative to healthy controls (Mukherjee et al., 2021; Nunomura et al., 2001; Sultana and Butterfield, 2024). In addition, the activity of major ROS-scavenging enzymes (e.g., SOD1, SOD2, glutathione peroxidase, catalase) is significantly reduced in the brains of AD patients, leading to a decline in ROS scavenging capacity and further exacerbating oxidative stress damage (Balendra and Singh, 2021; Guan et al., 2022). Notably, oxidative stress damage exhibits distinct synaptic specificity: the levels of oxidative stress markers in synaptosomes from the AD brain are much higher than those in mitochondrial or cytoplasmic fractions, indicating that synaptic sites are the primary targets of ROS attack (Naia et al., 2023).

A close vicious cycle exists between excessive mitochondrial ROS production and AD pathogenesis (Ionescu-Tucker and Cotman, 2021). Excessive-ROS-derived oxidative stress directly facilitates A β deposition and pathological tau aggregation to drive AD progression. Reciprocally, A β and aggregated tau further worsen mitochondrial dysfunction and boost ROS generation, establishing a vicious cycle between oxidative stress and AD pathogenesis. Regarding A β metabolism, ROS enhances A β production and impairs its clearance via multiple mechanisms. ROS induces oxidative modification of APP-processing enzymes including

β -secretase (BACE1) and γ -secretase. Oxidative stress elevates BACE1 expression and activity to enhance APP β -cleavage; meanwhile, ROS modifies γ -secretase subunits, increases APP-cleaving efficiency and consequently elevates A β output (Lee et al., 2021; Shaji et al., 2025). Second, ROS can inhibit A β clearance mechanisms, such as reducing the activity of A β -degrading enzymes including insulin-degrading enzyme and neprilysin in the brain, thus decreasing A β degradation. At the same time, ROS can damage the integrity of the blood-brain barrier, impairing the transport and clearance of A β to the periphery (Zuo et al., 2015). In addition, ROS can promote the aggregation and fibrillization of A β to form more toxic A β oligomers and fibrils, further exacerbating neuronal injury (Dahlgren et al., 2002). In terms of tau protein metabolism, ROS can promote the hyperphosphorylation of tau protein by activating various kinases, and simultaneously inhibit the activity of tau protein phosphatases, reducing the dephosphorylation of phosphorylated tau, which leads to the aggregation of hyperphosphorylated tau protein into neurofibrillary tangles (NFTs) (Xie et al., 2018). Furthermore, ROS can directly oxidatively modify tau protein, altering its conformation and function, enhancing its aggregation tendency, and reducing its binding capacity to microtubules, which disrupts microtubule stability, impairs axonal transport and cytoskeletal function, and ultimately causes neuronal dysfunction (Goodman et al., 2024; Nisha and Sarkar, 2021). Conversely, A β and abnormally aggregated tau protein can exacerbate mitochondrial dysfunction and promote ROS production through multiple pathways. A β can directly bind to mitochondria and localize to the inner mitochondrial membrane, disrupting the integrity of the electron transport chain, inhibiting the activity of OXPHOS complexes, leading to increased electron leakage and excessive ROS production (Cheignon et al., 2018; Kadowaki et al., 2005). Meanwhile, A β can induce the opening of the mitochondrial permeability transition pore (mPTP), disrupting the mitochondrial membrane potential, releasing pro-apoptotic factors such as cytochrome c, and further exacerbating ROS production (Angelova and Abramov, 2014; Trushina et al., 2022). Abnormally aggregated tau protein can bind to mitochondrial transport-related proteins (e.g., kinesin, dynein), inhibiting the axonal transport of mitochondria, leading to insufficient mitochondrial distribution at synaptic sites and an imbalance in energy supply, which in turn promotes ROS production. In addition, tau protein can activate neuro-inflammatory pathways, inducing microglia to release inflammatory factors, which indirectly damage mitochondrial function and exacerbate oxidative stress (Ising et al., 2019).

Excessive-ROS-triggered oxidative stress can also directly induce neuronal dysfunction and cell death. ROS oxidizes presynaptic neurotransmitter-vesicle proteins and postsynaptic receptor proteins, disrupting neurotransmitter release and binding and depressing synaptic transmission efficiency. Meanwhile, ROS disrupts the lipid bilayer of pre- and postsynaptic membranes, increasing synaptic-membrane permeability and triggering ion homeostasis disturbance, which ultimately gives rise to synaptic dysfunction and even synaptic loss (Li et al., 2026a; Wang et al., 2024a). In addition, ROS can oxidatively damage intracellular DNA (including nuclear DNA and mtDNA), leading to an increase in gene mutations and genomic instability, further exacerbating mitochondrial dysfunction and cellular damage. Meanwhile, ROS can activate the mitochondria-mediated endogenous apoptotic pathway, resulting in neuronal apoptosis (Abolhassani et al., 2017; Markesbery and Carney, 1999). Clinical studies have also provided evidence for the association between excessive ROS production and AD. Multiple studies have found that the levels of oxidative stress markers in the peripheral blood and cerebrospinal fluid of AD patients are significantly elevated, and these levels are negatively correlated with patients' cognitive function scores and positively correlated with disease duration, suggesting that oxidative stress markers may serve as potential biomarkers for the diagnosis and disease monitoring of AD (Coppede and Migliore, 2015; Knörrnschild et al., 2025). In addition, individuals carrying AD susceptibility genes (e.g., APOE ϵ 4) exhibit significantly higher ROS levels and more severe oxidative stress damage in brain and peripheral

cells than non-carriers, indicating that oxidative stress may be involved in the pathogenic process of AD susceptibility genes (Bhattarai et al., 2024; Jackson et al., 2024; Stevenson-Hoare et al., 2023).

To conclude, oxidative stress driven by excessive mitochondrial ROS constitutes a core node in AD pathogenesis. It forms a vicious cycle with A β deposition and pathological tau aggregation, which in turn induces neuronal dysfunction and death. Deciphering the regulatory mechanisms governing ROS generation-scavenging balance and the crosstalk between oxidative stress and AD pathological hallmarks is critical for developing oxidative-stress-targeted therapeutic strategies for AD.

2.4. Mitochondrial dynamics imbalance

Mitochondria are highly dynamic organelles that maintain their morphological, distributive and functional homeostasis through continuous fusion and fission processes, a dynamic balance known as mitochondrial dynamics (Al Amir Dache and Thierry, 2023). Mitochondrial fusion and fission are controlled by a set of highly conserved GTPases. Mitofusin 1 (MFN1) and Mitofusin 2 (MFN2) mediate outer-mitochondrial-membrane fusion, whereas Optic atrophy 1 (OPA1) governs inner-membrane fusion. Mitochondrial fission relies mainly on Dynamin-related protein 1 (DRP1), which is recruited from cytosol to mitochondria; Fission 1 (FIS1), Mitochondrial fission factor (MFF) and mitochondrial dynamics proteins of 49/51 kDa (MID49/51) modulate DRP1 localization and activation (Klemmensen et al., 2024; Roy et al., 2015). Under normal physiological conditions, the dynamic balance of mitochondrial fusion and fission is crucial for preserving the integrity of the mitochondrial network, facilitating the complementation and repair of mtDNA, OXPHOS function and eliminating damaged mitochondria (Bhatti et al., 2022).

Marked mitochondrial dynamics imbalance is observed in the brains of AD patients and AD model animals, primarily characterized by inhibited fusion and hyperactivated fission, which results in a fragmented mitochondrial morphology and severe functional impairment (Wasim, 2025). Multiple studies have confirmed abnormal mitochondrial morphology in neurons of the cerebral cortex and hippocampus in AD patients, featuring reduced volume, increased quantity, disorganized cristae structure and typical fragmentation characteristics (Sumpter et al., 1986; Zellmer et al., 2025). Studies have revealed a key mechanism in HT22 hippocampal neurons transfected with mutant amyloid precursor protein (mAPP): mAPP triggers excessive mitochondrial fragmentation by elevating mRNA and protein levels of fission-related DRP1 and FIS1 and repressing fusion-related MFN1, MFN2 and OPA1. Relative to control neurons, these model cells display increased mitochondrial number and shortened mitochondrial length, and abnormally enhanced DRP1 GTPase activity further aggravates mitochondrial structural disorganization. These findings first demonstrate that mAPP/A β -provoked mitochondrial dynamics imbalance preferentially takes place in hippocampal CA1 neurons. This observation matches the regional specificity of memory impairment in AD patients and offers a cellular basis for early-stage cognitive decline in AD (Kshirsagar et al., 2025; Reddy et al., 2024). At the molecular level, the expression levels of mitochondrial fusion-related proteins (MFN1, MFN2, OPA1) are significantly downregulated in the brains of AD patients, while those of fission-related proteins (DRP1, FIS1, MFF) are markedly upregulated, and such expression abnormalities are positively correlated with the pathological severity of AD (Singh et al., 2024). Studies have found that the mRNA and protein levels of MFN2 are significantly decreased in the hippocampus of AD patients, accompanied by a notable reduction in OPA1 protein levels and an increase in DRP1 protein levels. Hyperactivated DRP1 can enhance its mediated mitochondrial fission activity through post-translational phosphorylation modification, such as phosphorylation at the Ser616 site (Bera et al., 2022; Chen et al., 2024b; Pérez et al., 2017).

Mitochondrial dynamics imbalance leads to the failure of mitochondrial quality control through multiple mechanisms, thereby driving

the pathological progression of AD. First, reduced fusion causes impairment in mitochondrial functional complementation and mtDNA repair. Mitochondrial fusion facilitates the exchange of contents, including mtDNA, proteins and lipids, between different mitochondria, enabling functionally normal mitochondria to supply intact mtDNA and proteins to damaged ones, thus achieving mtDNA complementation and repair and maintaining the functional homeostasis of the mitochondrial population (Chen et al., 2010). The decreased expression of MFN1/2 and OPA1 in AD results in insufficient mitochondrial fusion, which blocks the mtDNA complementation and repair processes. Damaged mtDNA cannot be effectively repaired, leading to the gradual accumulation of mutation load and further exacerbation of OXPHOS dysfunction. Meanwhile, insufficient fusion disrupts the mitochondrial network, preventing functional coordination of mitochondria in different regions and further reducing the overall efficiency of energy metabolism (Cho et al., 2010; Dhapola et al., 2022). Second, excessive fission causes mitochondrial fragmentation and the accumulation of damaged mitochondria. Under normal conditions, moderate mitochondrial fission helps separate damaged mitochondria from the normal mitochondrial network, laying the foundation for the subsequent clearance of damaged mitochondria by mitophagy (Twig et al., 2008). However, excessive fission leads to extreme mitochondrial fragmentation, which impairs effective fusion and functional complementation of mitochondria and increases the proportion of damaged mitochondria (Ishihara et al., 2009). The hyperactivation of fission-related proteins such as DRP1 and FIS1 in AD results in uncontrolled mitochondrial fission. A large number of fragmented mitochondria suffer from functional impairment due to the inability to repair through fusion, and these damaged mitochondria are difficult to be effectively recognized and cleared by the autophagic system, leading to their massive accumulation in neurons and further aggravating oxidative stress and energy metabolism failure (Kim et al., 2017; Zhang et al., 2025).

In addition, mitochondrial dynamics imbalance impairs mitochondrial distribution and transport, exacerbating synaptic damage. Neurons have long axons and dendrites, and precise mitochondrial transport is required to deliver mitochondria to energy-consuming sites such as synapses. Mitochondrial morphology - tubular mitochondria in a fused state are more amenable to transport - is critical for transport efficiency (Lin and Sheng, 2015). The transport efficiency of excessively fragmented mitochondria is significantly reduced in AD, leading to insufficient mitochondrial number at synaptic sites, ATP shortage and increased ROS production. This directly impairs synaptic transmission and plasticity, resulting in synaptic dysfunction and even synaptic loss (Wang et al., 2020). Studies have shown a significant reduction in mitochondrial number in synaptosomes of AD model mice. Ameliorating mitochondrial dynamics balance by inhibiting DRP1 activity or over-expressing MFN2 can markedly increase mitochondrial number at synaptic sites, restore synaptic function and alleviate cognitive impairment in mice (Baek et al., 2017; Cai and Tammineni, 2017; Chen et al., 2024a). Moreover, mitochondrial dynamics imbalance is not an isolated mitochondrial dysfunction; it exhibits a tight bidirectional regulatory relationship with the core pathological features of AD (A β deposition and abnormal tau aggregation), and the two interact and promote each other to drive AD pathological progression. A β can directly regulate the expression and activity of mitochondrial dynamics-related proteins to induce dynamics imbalance. Studies have demonstrated that A β oligomers can bind to MFN2 and inhibit its mediated mitochondrial fusion activity (Park et al., 2015). Meanwhile, A β can activate calmodulin-dependent protein kinase II (CaMKII), leading to enhanced phosphorylation of DRP1 at the Ser616 site and promoting mitochondrial fission (Kandimalla et al., 2016). In addition, A β can indirectly downregulate the expression of fusion-related proteins and upregulate that of fission-related proteins by activating oxidative stress and neuroinflammatory pathways, further exacerbating mitochondrial fragmentation (Cho et al., 2009). Conversely, mitochondrial dynamics imbalance can promote A β production and tau hyperphosphorylation by

aggravating oxidative stress and energy metabolism failure. Fragmented mitochondria exhibit OXPHOS dysfunction and increased ROS production, which in turn activate the activity of BACE1 and γ -secretase to promote A β generation (Tong et al., 2005). At the same time, energy metabolism failure and oxidative stress can activate tau phosphorylation-related kinases and inhibit phosphatase activity, leading to tau hyperphosphorylation and aggregation (Farid et al., 2014).

Clinical studies have also provided evidence for the association between mitochondrial dynamics imbalance and AD. Multiple studies have found that mitochondria in peripheral blood leukocytes and skin fibroblasts of AD patients also present a fragmented morphology, with decreased expression of fusion-related proteins (MFN2, OPA1) and increased expression of the fission-related protein (DRP1), and these abnormalities are positively correlated with the degree of cognitive impairment in patients (Pakpian et al., 2020; Wang et al., 2012). In addition, individuals carrying the APOE ϵ 4 allele already exhibit abnormal expression of mitochondrial dynamics-related proteins in brain and peripheral cells, indicating that mitochondrial dynamics imbalance may be an early event in the pathogenesis of AD (Yin et al., 2020).

Collectively, mitochondrial dynamics imbalance constitutes a critical node for defective mitochondrial quality control in AD. By disrupting mitochondrial functional complementation, facilitating damaged-mitochondria accumulation and perturbing mitochondrial distribution and transport, it aggravates oxidative stress, energy metabolic failure and synaptic injury. It also establishes a vicious cycle with canonical AD pathological hallmarks and occupies a central position in AD progression. Therapeutic strategies aimed at restoring mitochondrial dynamics homeostasis represent a promising direction for AD treatment.

2.5. Mitochondrial mitophagy dysfunction

Mitophagy is a specialized form of cellular autophagy, defined as the process by which cells specifically recognize, encapsulate and degrade senescent, damaged or dysfunctional mitochondria through autophagic machinery. This process maintains the functional homeostasis of the mitochondrial population, eliminates intracellular oxidative stressors and recycles biological macromolecules (Jin and Youle, 2012; Li et al., 2026b). As a core component of the mitochondrial quality control system, mitophagy dysfunction leads to the massive accumulation of senescent/damaged mitochondria in cells, which in turn triggers oxidative stress, energy metabolism disorder and cellular injury - a process that plays a pivotal role in the pathological progression of AD (Kshirsagar et al., 2025; Skawratananond et al., 2025; Tran and Reddy, 2020).

The regulatory mechanism of mitophagy is complex, involving multiple signaling pathways and molecular events, among which the classic pathway mediated by PTEN-induced kinase 1 (PINK1) and Parkinson disease protein 2 (Parkin) is the most representative (Wang et al., 2019b). Under normal physiological conditions, in functionally intact mitochondria, PINK1 can be cleaved and degraded by proteases (e.g., PARL) on the inner mitochondrial membrane, maintaining a low level of expression. When mitochondria are damaged (e.g., with decreased membrane potential), PINK1 cannot be effectively cleaved, thus accumulating and being activated on the outer membrane of damaged mitochondria. It then recruits cytoplasmic Parkin, an E3 ubiquitin ligase, to the mitochondrial surface (Meissner et al., 2011). Once activated, Parkin mediates the ubiquitination modification of various mitochondrial surface proteins (e.g., MFN1/2, VDAC1, Tom20). These ubiquitinated proteins serve as recognition signals for autophagic receptors, recruiting autophagy-related proteins to form autophagosomes. Eventually, autophagosomes fuse with lysosomes to degrade damaged mitochondria (Lazarou et al., 2015). In addition, PINK1-Parkin-independent mitophagic pathways exist, such as the direct mediation of mitochondria-autophagosome binding by autophagic receptors (e.g., BNIP3, NIX, FUNDC1). These receptor proteins can bind to both mitochondrial membrane proteins and

Microtubule-associated protein 1 light chain 3 (LC3), enabling the specific recognition and degradation of damaged mitochondria (Schweers et al., 2007).

Marked mitophagy dysfunction is observed in the brains of AD patients and AD model animals, mainly characterized by reduced mitophagic activity that fails to efficiently clear senescent/damaged mitochondria, leading to their massive accumulation in neurons (Fang et al., 2019). Multiple studies have confirmed a significant increase in the number of damaged mitochondria (with abnormal morphology and OXPHOS dysfunction) in neurons of key brain regions such as the cerebral cortex and hippocampus in AD patients. Meanwhile, the binding efficiency of autophagosomes to mitochondria and the fusion efficiency of autophagosomes with lysosomes are significantly decreased, indicating a block in the mitophagic process (Fang et al., 2019; Uddin et al., 2019). At the molecular level, the expression and activity of key molecules in the PINK1-Parkin-mediated mitophagic pathway are markedly abnormal in the AD brain: the mRNA and protein levels of PINK1 and Parkin are significantly downregulated, preventing the effective recognition and labeling of damaged mitochondria. Additionally, the expression levels of autophagic receptors (e.g., BNIP3, FUNDC1) are also notably reduced, impairing the binding of autophagosomes to damaged mitochondria (Pinjala et al., 2024; Zhou et al., 2023). Furthermore, lysosomal dysfunction (e.g., decreased lysosomal enzyme activity, increased lysosomal membrane permeability) blocks the final degradation step of mitophagy (Zhang et al., 2021c). Studies have found that the protein levels of PINK1 and Parkin are significantly decreased in the hippocampus of AD patients, while the LC3-II/LC3-I ratio, a marker of autophagic activity, is markedly reduced. Concurrently, the activity of lysosomal cathepsin D is significantly diminished, suggesting abnormalities in the initiation, execution and degradation stages of mitophagy (Hou et al., 2020; Qi et al., 2025).

This multi-stage mitophagy dysfunction is not merely a manifestation of mitochondrial disorder during AD progression; it exerts cross-regulation with the core pathological changes of AD through multiple molecular mechanisms, directly driving disease pathogenesis. Damaged mitochondria exhibit OXPHOS dysfunction and increased electron leakage, leading to massive ROS production. Moreover, these damaged mitochondria cannot be timely cleared by the autophagic system and continuously generate ROS, triggering severe oxidative stress responses that damage intracellular biological macromolecules (Patro et al., 2021). In addition, the energy production capacity of damaged mitochondria is significantly impaired, and their massive accumulation results in insufficient global ATP supply in neurons, exacerbating energy metabolism failure, impairing energy-consuming processes such as synaptic transmission and axonal transport, and ultimately causing neuronal dysfunction (Du et al., 2010). Second, mitophagy dysfunction promotes A β deposition and abnormal tau aggregation (Fang et al., 2019). On the one hand, excessive ROS generated by damaged mitochondria activates the activity of BACE1 and γ -secretase to promote A β production, while inhibiting A β degradation and clearance, leading to A β deposition (Tong et al., 2005). On the other hand, oxidative stress and energy metabolism failure activate tau phosphorylation-related kinases (e.g., GSK3 β , CDK5) and inhibit phosphatase activity, resulting in tau hyperphosphorylation and aggregation (Bai et al., 2022). Furthermore, mitophagy dysfunction can promote A β production by interfering with the metabolic process of APP: studies have shown that autophagosomes are involved in APP degradation, and impaired mitophagy leads to intracellular APP accumulation, increasing the probability of its cleavage by β - and γ -secretase and thus promoting A β generation (Zhang et al., 2017). Third, mitophagy dysfunction exacerbates neuronal apoptosis. Damaged mitochondria not only generate excessive ROS but also release pro-apoptotic factors such as cytochrome c and Smac/-DIABLO, activating the endogenous apoptotic pathway. Mitophagy dysfunction prevents the clearance of these damaged mitochondria, which continuously release pro-apoptotic factors and ultimately induce neuronal apoptosis (Meng et al., 2025).

Notably, mitophagic dysfunction and canonical AD pathological hallmarks do not act in a unidirectional manner; rather, they undergo tight bidirectional crosstalk to create a self-reinforcing pathological cycle that amplifies brain injury in AD. As key AD pathological mediators, A β and pathologically aggregated tau directly suppress mitophagy via multiple molecular cascades. A β oligomers bind PINK1 and block its accumulation and activation on damaged-mitochondrial surfaces, thereby abrogating PINK1-Parkin-dependent mitophagy. Meanwhile, A β activates the mTOR pathway to repress autophagy initiation (Li et al., 2022). Hyperphosphorylated tau protein can bind to autophagy-related proteins, inhibiting the formation and maturation of autophagosomes. Additionally, NFTs formed by tau aggregation can physically hinder the fusion of autophagosomes with lysosomes, further suppressing mitophagy (Heckmann et al., 2019). Conversely, the accumulation of damaged mitochondria caused by reduced mitophagic activity further promotes A β production and abnormal tau aggregation by exacerbating oxidative stress and energy metabolism failure, thus forming a vicious cycle (Fang et al., 2019). Clinical studies have also provided evidence for the association between mitophagy dysfunction and AD. Multiple studies have found significantly abnormal levels of autophagy-related markers (e.g., LC3-II, Beclin1, Parkin) in the cerebrospinal fluid of AD patients, which correlate with patients' cognitive function scores and disease duration, suggesting that these markers may serve as potential diagnostic tools for AD (Atya et al., 2024; Salminen et al., 2013). In addition, individuals carrying AD susceptibility genes (e.g., APOE ϵ 4, PSEN1) already exhibit reduced mitophagic activity in brain and peripheral cells, indicating that mitophagy dysfunction may be an early event in AD pathogenesis (Alkhalifa et al., 2023; Karch and Goate, 2015).

To integrate, mitophagy dysfunction is the core mechanism underlying the accumulation of senescent/damaged mitochondria in AD. It plays a pivotal role in AD pathological progression by exacerbating oxidative stress, energy metabolism disorder, A β deposition, abnormal tau aggregation and neuronal apoptosis. An in-depth elucidation of the molecular mechanisms of mitophagy dysfunction and its interaction with AD pathological features is of great significance for the development of AD intervention strategies targeting mitophagy.

2.6. Mitochondrial Ca²⁺ imbalance & intrinsic apoptosis overactivation

Calcium (Ca²⁺) is an important intracellular second messenger that participates in the regulation of various physiological processes including neurotransmitter release, synaptic plasticity, gene expression, cell proliferation and apoptosis (Terrell et al., 2023). Intraneuronal Ca²⁺ homeostasis is crucial for maintaining normal neuronal function, and mitochondria play a central role in the regulation of Ca²⁺ homeostasis. Mitochondria take up cytoplasmic Ca²⁺ through the mitochondrial calcium uniporter (MCU) on the inner mitochondrial membrane and release Ca²⁺ back into the cytoplasm via the mitochondrial sodium-calcium exchanger (NCLX) and rapid mitochondrial calcium uptake mechanisms, thus forming a dynamic Ca²⁺ balance between mitochondria and the cytoplasm (Kashif et al., 2023; Liu et al., 2017). Under normal physiological conditions, the Ca²⁺ uptake function of mitochondria not only regulates cytoplasmic Ca²⁺ concentration to avoid cellular damage caused by excessive Ca²⁺ accumulation, but also activates dehydrogenases in the mitochondrial matrix (e.g., pyruvate dehydrogenase, isocitrate dehydrogenase) in a Ca²⁺-dependent manner, thereby promoting the tricarboxylic acid cycle, OXPHOS function and increasing ATP production (Gowda et al., 2022).

However, marked mitochondrial Ca²⁺ homeostasis imbalance is observed in the brains of AD patients and AD model animals, mainly characterized by increased mitochondrial Ca²⁺ uptake and decreased release, which leads to excessive mitochondrial Ca²⁺ accumulation and a sustained elevation of cytoplasmic Ca²⁺ concentration, disrupting intracellular Ca²⁺ homeostasis (Reddy et al., 2005). Multiple studies have confirmed that the Ca²⁺ content is significantly increased in

mitochondria of neurons in the cerebral cortex and hippocampus of AD patients, and the cytoplasmic Ca²⁺ concentration is markedly higher in AD patients than in healthy controls both at rest and after stimulation (Chaudhary et al., 2025; Joshi et al., 2023). At the molecular level, the expression and activity of proteins related to mitochondrial Ca²⁺ homeostasis regulation are significantly abnormal in AD: the expression and activity of MCU are markedly upregulated, promoting mitochondrial uptake of cytoplasmic Ca²⁺; the expression and activity of NCLX are significantly downregulated, inhibiting mitochondrial Ca²⁺ release; meanwhile, the number of endoplasmic reticulum-mitochondrial contact sites (MAMs) is increased, facilitating the transfer of endoplasmic reticulum Ca²⁺ to mitochondria (Calvo-Rodriguez and Bacskaï, 2020; Hedskog et al., 2013; Jadia et al., 2019). For example, the protein level of MCU is significantly elevated and that of NCLX is markedly decreased in the hippocampus of AD patients, while the expression levels of MAMs-related proteins (e.g., IP3R, VDAC1, Grp75) are significantly upregulated. This causes efficient transfer of endoplasmic reticulum Ca²⁺ to mitochondria through MAMs, further exacerbating mitochondrial Ca²⁺ accumulation (Area-Gomez and Schon, 2017b; Cai et al., 2022; Jadia et al., 2019).

Mitochondrial Ca²⁺ homeostasis disruption triggers neuronal loss via activating the intrinsic apoptotic pathway, representing one critical node in AD pathogenesis. Excessive mitochondrial Ca²⁺ overload promotes mPTP opening. As a multi-protein non-selective channel residing on the inner mitochondrial membrane, mPTP stays closed under physiological conditions and opens in response to mitochondrial Ca²⁺ overloading, aggravated oxidative stress or declined mitochondrial membrane potential (Bonora et al., 2022). The opening of mPTP results in a significant increase in the permeability of the inner mitochondrial membrane, the disappearance of the transmembrane proton gradient, the collapse of mitochondrial membrane potential and the complete loss of OXPHOS function, leading to the cessation of ATP production. Meanwhile, the increased osmotic pressure in the mitochondrial matrix causes mitochondrial swelling and outer membrane rupture, releasing mitochondrial pro-apoptotic factors such as cytochrome c, Smac/DIABLO and apoptosis-inducing factor (AIF) (Bernardi et al., 2023). After being released into the cytoplasm, cytochrome c binds to apoptotic protease-activating factor 1 and procaspase-9 to form an apoptosome, which activates caspase-9, an initiator caspase, and then further activates downstream effector caspases (e.g., caspase-3, caspase-7). This triggers a cascade reaction that leads to the degradation of key intracellular proteins and ultimately results in neuronal apoptosis (Budihardjo et al., 1999; Li et al., 1997). Following its release, Smac/DIABLO binds to cytoplasmic inhibitor of apoptosis proteins (IAPs), relieving the inhibitory effect of IAPs on caspases and further promoting the activation of the apoptotic pathway (Du et al., 2000). After being released, AIF can directly enter the nucleus, causing chromatin condensation and DNA fragmentation, and inducing caspase-independent neuronal apoptosis (Joza et al., 2001). In addition, mitochondrial Ca²⁺ homeostasis imbalance can indirectly promote neuronal apoptosis by exacerbating oxidative stress and energy metabolism failure. Excessive mitochondrial Ca²⁺ accumulation can cause OXPHOS dysfunction and increased electron leakage, leading to excessive ROS production. At the same time, Ca²⁺ can activate cytoplasmic Ca²⁺-dependent proteases (e.g., calpains), which degrade intracellular antioxidant enzymes (e.g., SOD, glutathione peroxidase), further aggravating oxidative stress damage (Duchen, 2012). Oxidative stress and energy metabolism failure can further activate the apoptotic pathway through multiple approaches such as DNA damage, cytoskeletal disruption and impaired gene expression, ultimately leading to neuronal apoptosis (Valero, 2014).

There is a close interaction between mitochondrial Ca²⁺ homeostasis imbalance and the pathological features of AD. A β and abnormally aggregated tau protein can disrupt mitochondrial Ca²⁺ homeostasis through multiple pathways: A β oligomers can bind to neuronal membrane Ca²⁺ channels (e.g., NMDA receptors, voltage-gated calcium

channels) and activate their opening, leading to elevated cytoplasmic Ca^{2+} concentration. Meanwhile, A β can bind to MAMs-related proteins, promoting the transfer of endoplasmic reticulum Ca^{2+} to mitochondria and exacerbating mitochondrial Ca^{2+} accumulation (Sanz-Blasco et al., 2008). Hyperphosphorylated tau protein can bind to mitochondrial membrane proteins (e.g., VDAC1), impairing the mitochondrial Ca^{2+} uptake and release functions. Additionally, tau aggregation can disrupt the cytoskeleton, affecting the distribution and function of Ca^{2+} pumps (e.g., plasma membrane Ca^{2+} -ATPase, sarco/endoplasmic reticulum Ca^{2+} -ATPase), resulting in impaired cytoplasmic Ca^{2+} clearance and further exacerbating Ca^{2+} homeostasis imbalance (Britti et al., 2020). Conversely, oxidative stress, energy metabolism failure and apoptotic pathway activation induced by mitochondrial Ca^{2+} homeostasis imbalance can promote A β production and tau hyperphosphorylation, forming a vicious cycle. Clinical studies have also provided evidence for the association between mitochondrial Ca^{2+} homeostasis imbalance and AD. Multiple studies have found that the Ca^{2+} concentration in peripheral blood leukocytes and cerebrospinal fluid of AD patients is significantly elevated, and this elevation is positively correlated with the degree of cognitive impairment and the proportion of neuronal loss in patients (Adunsky et al., 1991; Hartmann et al., 1994). In addition, individuals carrying AD susceptibility genes (e.g., APOE ϵ 4, PSEN1) already exhibit abnormalities in neuronal Ca^{2+} homeostasis regulation, characterized by elevated cytoplasmic Ca^{2+} concentration and increased mitochondrial Ca^{2+} uptake, indicating that Ca^{2+} homeostasis imbalance may be an early event in AD pathogenesis.

Taken together, mitochondrial Ca^{2+} homeostasis imbalance directly leads to neuronal loss by activating the intrinsic apoptotic pathway and forms a vicious cycle with the classic pathological features of AD, playing a pivotal role in the pathological progression of AD. An in-depth elucidation of the molecular mechanisms of mitochondrial Ca^{2+} homeostasis imbalance and its association with the apoptotic pathway is of great significance for the development of AD intervention strategies targeting Ca^{2+} homeostasis regulation.

2.7. Mitochondrial transport defects

Neurons are highly polarized cells with long axons and dendrites. Synapses, the key structures for information transmission between neurons, are typically located at axon terminals or dendritic spines. Since synaptic transmission consumes a large amount of ATP and requires the timely clearance of harmful substances such as ROS, sufficient and functionally intact mitochondria are essential to support synaptic sites (Mattson et al., 2008). Through anterograde transport and retrograde transport, mitochondria are precisely distributed to energy-consuming sites such as synapses, and senescent/damaged mitochondria are transported back to the soma for repair or degradation. Mitochondrial transport is a highly complex process, which mainly takes microtubules as the transport track, with kinesins mediating anterograde transport and dyneins mediating retrograde transport. Meanwhile, a variety of regulatory proteins (e.g., Milton, Miro, Trak1/2) are involved in regulating the binding of mitochondria to motor proteins, as well as the direction and speed of transport (Saxton and Hollenbeck, 2012).

Marked mitochondrial transport defects are observed in the brains of AD patients and AD model animals, mainly characterized by reduced anterograde transport and impaired retrograde transport. This leads to an insufficient number of mitochondria at synaptic sites, and senescent/damaged mitochondria cannot be timely transported back to the soma for clearance, thereby causing synaptic dysfunction and abnormal neural signal transmission (Calkins and Reddy, 2011). Multiple studies have confirmed that in the axons of neurons in the cerebral cortex and hippocampus of AD patients, the transport speed of mitochondria is significantly reduced, the proportion of mitochondria undergoing anterograde transport is markedly decreased, and the number of mitochondria in synaptosomes is drastically reduced. Moreover, most of

these mitochondria exhibit abnormal morphology and OXPHOS dysfunction (Calkins and Reddy, 2011; Flannery and Trushina, 2019). At the molecular level, proteins governing mitochondrial transport exhibit prominent expression and functional abnormalities in AD. Kinesins (e.g., KIF5A, KIF5B) are markedly down-regulated with impaired mitochondrial-binding capacity. Regulatory factors including Miro1 and Milton show reduced expression, accompanied by oxidative modification and degradation. In addition, tau-hyperphosphorylation-triggered microtubule defects, such as enhanced depolymerization and diminished stability, also disrupt mitochondrial transport tracks (Grundke-Iqbal et al., 1986; Wang et al., 2019a). For example, the protein levels of KIF5A and Miro1 are significantly reduced in the hippocampus of AD patients, while hyperphosphorylated tau protein can bind to microtubules, disrupting their integrity and stability and thus damaging the mitochondrial transport track (Wang et al., 2019a; Zeng et al., 2025).

Mitochondrial transport defects cause synaptic dysfunction and abnormal neural signal transmission through multiple mechanisms, thereby driving the pathological progression of AD. First, an insufficient number of mitochondria at synaptic sites leads to energy supply shortage. Synaptic transmission processes (including the synthesis, release and reuptake of neurotransmitters, and the opening and closing of ion channels on the postsynaptic membrane) require a large amount of ATP. Anterograde transport defects in AD result in a significant reduction in the number of functionally intact mitochondria at synaptic sites and insufficient ATP production, which directly impairs the efficiency of synaptic transmission (Flannery and Trushina, 2019). Studies have shown that ATP levels in synaptosomes of AD model mice are markedly decreased, and supplementation with exogenous ATP or an increase in the number of synaptic mitochondria by improving mitochondrial transport can significantly restore synaptic transmission function (Pradeepkiran et al., 2024; Reddy, 2007). In addition, energy supply shortage also affects the synthesis and modification of synaptic plasticity-related proteins (e.g., brain-derived neurotrophic factor, postsynaptic density protein 95), leading to impaired synaptic plasticity and thus compromising learning and memory functions (Cai and Tammineni, 2017). Second, the accumulation of damaged mitochondria at synaptic sites exacerbates oxidative stress damage (Du et al., 2010). Retrograde transport defects in AD prevent senescent/damaged mitochondria from being timely transported back to the soma for repair or degradation, resulting in their massive accumulation at synaptic sites. These damaged mitochondria exhibit OXPHOS dysfunction and increased ROS production, while the antioxidant reserves at synaptic sites are limited. Excessive ROS causes oxidative damage to neurotransmitter vesicle proteins on the presynaptic membrane, receptor proteins on the postsynaptic membrane and neurotransmitters in the synaptic cleft, leading to reduced neurotransmitter release and decreased receptor sensitivity, which in turn impairs neural signal transmission (Seelert et al., 2009). For instance, ROS can oxidatively modify synaptic vesicle protein 2 (SV2) on the presynaptic membrane to inhibit neurotransmitter release (Ștefănescu et al., 2019); meanwhile, ROS can cause oxidative damage to NMDA receptors on the postsynaptic membrane, reducing their sensitivity to glutamate and impairing the generation of postsynaptic potentials (Simonyi et al., 2010). Third, mitochondrial transport defects affect the stability of synaptic structures. The maintenance and remodeling of synapses require continuous energy supply and material transport. Energy shortage and oxidative stress caused by mitochondrial transport defects impair the structural integrity of the presynaptic and postsynaptic membranes, leading to synaptic atrophy, widened synaptic clefts and even synaptic loss (Li et al., 2026b). A study has found a significant reduction in synaptic density in the brains of AD patients, and the degree of synaptic loss is positively correlated with the severity of cognitive impairment. Notably, regions with synaptic loss are often accompanied by mitochondrial transport defects and mitochondrial dysfunction (Mehder et al., 2020).

There is a close interaction between mitochondrial transport defects

and the pathological features of AD. A β and abnormally aggregated tau protein are important factors leading to mitochondrial transport defects: A β oligomers can bind to Miro1 and promote its ubiquitin-dependent degradation, thereby disrupting the binding of mitochondria to kinesins and inhibiting anterograde transport (Vaillant-Beuchot et al., 2024); meanwhile, A β can activate oxidative stress pathways to damage microtubule structures and impair the mitochondrial transport track (Stowers et al., 2002). Hyperphosphorylated tau protein can bind to microtubules, disrupting their stability and competing with kinesins for microtubule binding, thus inhibiting kinesin-mediated mitochondrial anterograde transport (Iqbal et al., 1998). Conversely, synaptic dysfunction and oxidative stress caused by mitochondrial transport defects can further exacerbate AD pathological progression by promoting A β production and tau hyperphosphorylation, forming a vicious cycle (Pradeepkiran and Reddy, 2020). Clinical studies have also provided evidence for the association between mitochondrial transport defects and AD. Multiple studies have found that the levels of mitochondrial transport-related proteins (e.g., KIF5A, Miro1) in the cerebrospinal fluid of AD patients are significantly reduced, and these levels correlate with patients' cognitive function scores and the degree of synaptic dysfunction, suggesting that these proteins may serve as potential biomarkers for AD (Veverová et al., 2024). In addition, individuals carrying the APOE4 allele already exhibit abnormal mitochondrial transport function in neurons, characterized by reduced transport speed and a decreased proportion of anterograde transport, indicating that mitochondrial transport defects may be an early event in AD pathogenesis (Jackson et al., 2024).

In brief, mitochondrial-transport defects contribute substantially to synaptic dysfunction and aberrant neuronal signal transduction. They drive AD pathogenesis by triggering energy insufficiency, oxidative injury and synaptic structural disruption. Deciphering the molecular mechanisms underlying such defects and their crosstalk with AD pathological hallmarks is critical for developing mitochondria-transport-targeted AD interventions.

2.8. Bidirectional regulatory network between tau pathology and mitochondrial dysfunction

Although A β deposition has long been regarded as the core pathological hallmark of AD, accumulating clinical and preclinical evidence indicates that the direct crosstalk between abnormal aggregation of pathological tau protein and mitochondrial dysfunction is a more critical and direct driver of progressive neuronal damage and cognitive decline in AD (Quntanilla and Tapia-Monsalves, 2020; Reddy, 2011). Unlike A β , which exerts its toxicity primarily by binding to respiratory chain complexes on the inner mitochondrial membrane, pathological tau can invade mitochondria via multiple pathways and disrupt all aspects of mitochondrial function. Concurrently, mitochondrial dysfunction in turn promotes the hyperphosphorylation, aberrant cleavage, and aggregation of tau (Briston and Hicks, 2018). This mechanism not only explains the limited clinical efficacy of A β -targeted therapies but also provides a novel core target for the early intervention of AD. Pathological tau exists in three major forms: monomers, soluble oligomers, and insoluble NFTs. Among them, soluble tau oligomers exhibit the most potent mitochondrial toxicity and appear prior to the formation of NFTs, serving as the primary trigger of early mitochondrial dysfunction in AD (Lasagna-Reeves et al., 2011). Hyperphosphorylated tau can enter mitochondria through the translocase of the outer membrane (TOM)/translocase of the inner membrane (TIM) complex or directly bind to mitochondrial membrane proteins, causing multilayered damage to all core physiological functions of mitochondria (Olesen and Quintanilla, 2023).

First, oxidative phosphorylation impairment occurs: hyperphosphorylated tau directly binds to VDAC1 on the outer mitochondrial membrane and adenine nucleotide translocase (ANT) on the inner mitochondrial membrane, disrupts inner mitochondrial membrane

permeability and the transmembrane proton gradient, and reduces ATP synthesis efficiency (Shoshan-Barmatz et al., 2018). Recent structural biology studies have confirmed that tau oligomers specifically insert into the hydrophilic arm domain of mitochondrial complex I to block electron transport, with an inhibitory effect significantly stronger than that of A β oligomers. This inhibition is most prominent in hippocampal CA1 neurons, which is highly consistent with the brain region specificity of memory impairment in AD patients (Oliver and Reddy, 2019a). In addition, tau further exacerbates the assembly defects of respiratory chain complexes by inhibiting the transport of nuclear-encoded mitochondrial proteins (Devi et al., 2006). Second, mitochondrial dynamics imbalance develops: pathological tau disrupts the dynamic balance of mitochondrial fusion and fission through a dual mechanism. On the one hand, tau activates GSK3 β and CDK5, promotes the phosphorylation of DRP1 at the Ser616 site, and enhances its mediated mitochondrial fission activity (Kandimalla et al., 2016). On the other hand, tau downregulates the expression level of MFN2 via the ubiquitin-proteasome pathway (Abtahi et al., 2020). Most critically, tau oligomers directly bind to OPA1 and disrupt its mediated inner mitochondrial membrane fusion, which is the core mechanism underlying tau-induced synaptic mitochondrial dysfunction and synaptic loss (Pérez et al., 2018). Furthermore, tau pathology blocks the complete process of mitophagy at multiple stages: soluble tau oligomers directly bind to autophagy-related proteins Beclin1 and LC3, inhibiting the formation and maturation of autophagosomes; neurofibrillary tangles block the fusion of autophagosomes and lysosomes through physical spatial obstruction (Kang et al., 2019). Meanwhile, tau inhibits the mitochondrial translocation and activation of PTEN-PINK1, blocks the PINK1-Parkin-mediated mitophagy pathway, and leads to the massive accumulation of damaged mitochondria in neurons (Lyu and Li, 2026). In addition, pathological tau directly binds to the MCU to enhance mitochondrial uptake of cytoplasmic calcium, while inhibiting the calcium release function of the NCLX, resulting in mitochondrial calcium overload (Britti et al., 2020). This calcium overload not only induces the opening of the mPTP and the release of pro-apoptotic factors such as cytochrome c to trigger neuronal apoptosis, but also further activates calcium-dependent kinases to promote tau hyperphosphorylation, forming a local vicious cycle of “tau-calcium overload-tau phosphorylation” (Calvo-Rodríguez and Bacskaï, 2020). Notably, the characteristic impairment of mitochondrial transport by tau is the main cause of early synaptic dysfunction in AD. Hyperphosphorylated tau binds to tubulin, disrupts the stability and integrity of microtubules, and competitively inhibits the binding of KIF5A to microtubules, leading to impaired anterograde transport of mitochondria (Prots et al., 2013). A study has shown that tau-induced mitochondrial transport defects result in a significant reduction in the number of functional mitochondria at synaptic sites, causing insufficient synaptic energy supply and oxidative stress damage, and ultimately leading to synaptic dysfunction and cognitive decline (Du et al., 2010). Numerous clinical studies have confirmed that the distribution range and load of tau pathology in the brain have a stronger correlation with the degree and progression rate of cognitive impairment in AD patients compared with A β pathology (Jack et al., 2013; Weber et al., 2026; Wuestefeld et al., 2025). Therefore, intervention strategies targeting tau-mitochondrial crosstalk have become a research hotspot for the early treatment of AD. Currently, multiple drugs targeting this pathway have shown significant neuroprotective effects in preclinical studies: for example, peptide P110, which specifically blocks the interaction between tau and DRP1, effectively inhibits excessive mitochondrial fragmentation and improves cognitive function (Qi et al., 2013); small molecule inhibitors of tau mitochondrial translocation reduce tau accumulation in mitochondria and protect mitochondrial respiratory function (Pradeepkiran et al., 2021). In addition, combined therapeutic strategies targeting both tau pathology and mitochondrial dysfunction are expected to achieve better efficacy than single-target therapies.

Taken together, there exists a tight bidirectional regulatory

relationship between tau pathology and mitochondrial dysfunction, and this mechanism serves as a core amplifier of Alzheimer's disease (AD) pathological progression. Tau-mitochondrial crosstalk not only participates in the initiation and progression of core mitochondrial functional abnormalities, but also exhibits complex crosstalk with other AD pathological processes such as A β pathology and neuroinflammation, collectively constituting the intricate pathological network of AD.

3. Mitochondrial functional interaction network

Mitochondrial dysfunction in AD is not manifested as an isolated abnormality of a single mechanism, but an intricate interconnected regulatory network formed by the mutual interaction and cross-talk among the aforementioned core mechanisms, which collectively drive the pathological progression of AD (Oliver and Reddy, 2019a). Through cascade amplification effects, synergistic regulatory actions and bidirectional crosstalk with the AD microenvironment, these mechanisms constitute the core regulatory network underlying the pathological progression of AD. An in-depth dissection of this network is crucial for a comprehensive understanding of the pathogenesis of AD.

3.1. mtDNA-oxidative stress-mitochondrial dynamics cascade

A distinct cascade amplification effect exists among mtDNA mutations, oxidative stress and mitochondrial dynamics imbalance. MtDNA mutations exacerbate oxidative stress, and the resultant dynamics imbalance further aggravates mtDNA damage, which acts as a core driving force for the persistent deterioration of mitochondrial function

(Fig. 3). To fully understand this pathology, it is essential to elucidate the specific mechanisms of cascade amplification between mitochondrial dysfunction and amyloid pathology. A classic *in vivo* example is the vicious cycle driven by the interaction between A β and mitochondrial proteins. Initial mild mitochondrial electron leakage produces ROS, which upregulates the expression and activity of BACE1, thereby promoting excessive A β generation. Subsequently, A β oligomers actively translocate into mitochondria and directly bind to amyloid-binding alcohol dehydrogenase (ABAD) (Lustbader et al., 2004). This A β -ABAD interaction severely disrupts mitochondrial membrane permeability and Complex IV activity, exacerbating electron leakage and further amplifying ROS production. This catastrophic cascade directly transforms localized metabolic stress into global bioenergetic failure, driving early synaptic toxicity in AD (De Strooper and Karran, 2016).

MtDNA mutations serve as the initiating step of this cascade reaction. As mentioned previously, mtDNA lacks histone protection and has an underdeveloped repair system, rendering it susceptible to ROS-induced damage and subsequent mutations. MtDNA mutations directly reduce the activity of OXPHOS complexes, block the electron transport chain and increase electron leakage, thereby promoting ROS production (Santos et al., 2013). Meanwhile, mtDNA mutations impair mitochondrial protein synthesis, leading to insufficient assembly of OXPHOS complexes, which further exacerbates electron transport chain dysfunction and ROS generation, thus triggering oxidative stress responses (Taylor and Turnbull, 2005). Oxidative stress acts as a key intermediate link connecting mtDNA mutations and mitochondrial dynamics imbalance. Excessive ROS not only directly induces oxidative

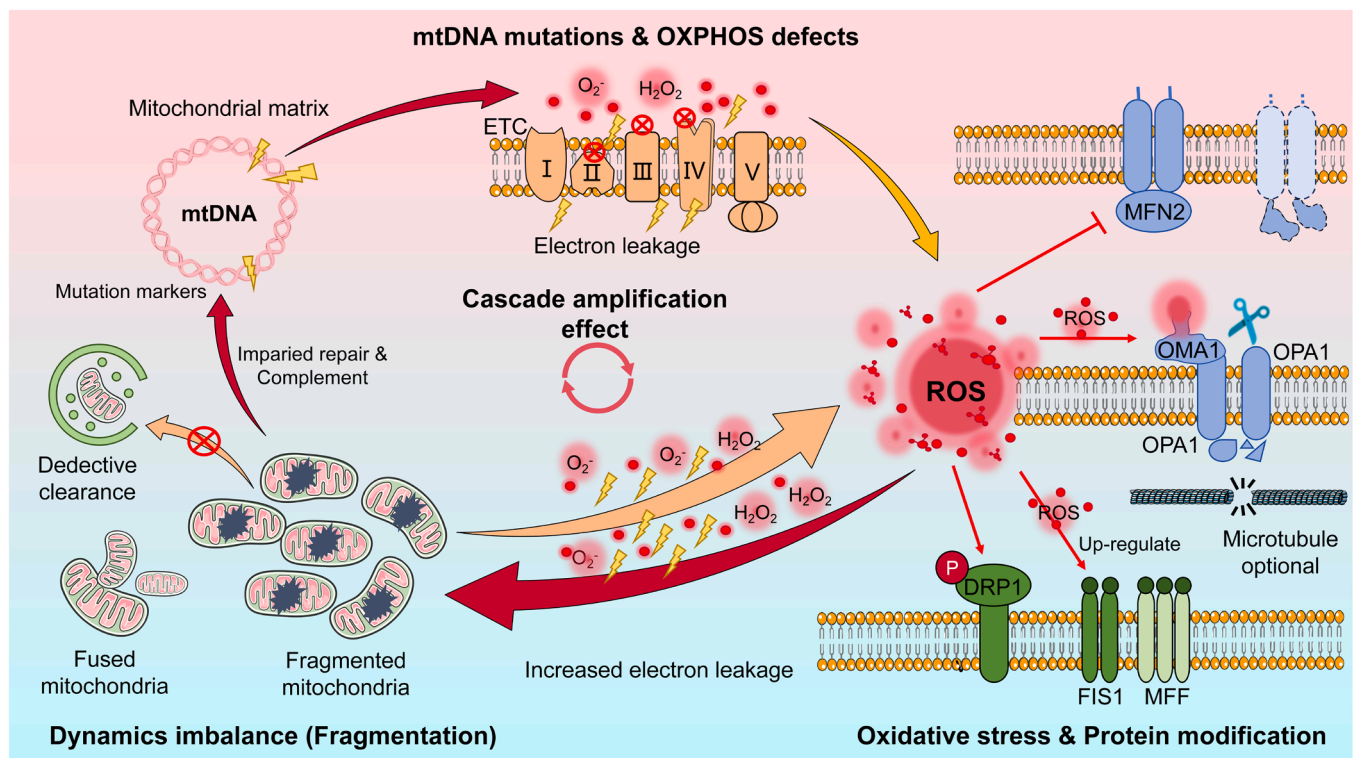


Fig. 3. ROS amplification cascade and mitochondrial dynamics imbalance driven by mtDNA mutations and OXPHOS defects. mtDNA mutations and OXPHOS defects trigger electron leakage from the ETC, leading to the generation of ROS including $O_2^{\cdot-}$ and H_2O_2 . Elevated ROS levels induce oxidative stress and protein modifications, which disrupt mitochondrial dynamics: ROS inhibits MFN2 to suppress mitochondrial fusion, activates metallopeptidase OMA1 to cleave OPA1, and upregulates DRP1 along with its adaptors FIS1 and MFF to promote mitochondrial fission. This shift toward excessive fission results in mitochondrial fragmentation, which further increases electron leakage from the ETC, creating a self-amplifying cascade of ROS production. Concurrently, fragmented mitochondria exhibit defective clearance, impairing mtDNA repair and complementation, which perpetuates mtDNA mutations and OXPHOS dysfunction, forming a vicious cycle that drives pathological progression. mtDNA (mitochondrial DNA), OXPHOS (oxidative phosphorylation), ETC (electron transport chain), ROS (reactive oxygen species), $O_2^{\cdot-}$ (superoxide anion), H_2O_2 (hydrogen peroxide), MFN2 (mitofusin 2), OMA1 (metallopeptidase OMA1), OPA1 (optic atrophy 1), DRP1 (dynamamin-related protein 1), FIS1 (fission 1), MFF (mitochondrial fission factor). Figure created with PowerPoint.

damage to mtDNA, leading to more mtDNA mutations, but also disrupts the dynamic balance of mitochondrial fusion and fission by modifying mitochondria dynamics-related proteins (Sena and Chandel, 2012). On the one hand, ROS can oxidatively modify fusion-related proteins (e.g., MFN1, MFN2, OPA1), resulting in conformational changes, reduced activity and even degradation of these proteins. Studies have found that ROS can inhibit MFN2-mediated outer mitochondrial membrane fusion activity by oxidizing the cysteine residues of MFN2; meanwhile, ROS can activate proteases (e.g., OMA1), promoting the cleavage and degradation of OPA1 and thus inhibiting inner mitochondrial membrane fusion (Baker et al., 2014). On the other hand, ROS can activate the phosphorylation modification of the fission-related protein DRP1, enhancing its mediated mitochondrial fission activity; additionally, ROS can upregulate the expression of fission-related proteins such as FIS1 and MFF, further promoting mitochondrial fission (Chen et al., 2021). Furthermore, ROS disrupt microtubule architecture, disturb mitochondrial morphology, and indirectly worsen mitochondrial dynamics imbalance. Such imbalance in turn amplifies injury triggered by mtDNA mutations and oxidative stress. Impaired fusion blocks inter-mitochondrial mtDNA complementation and repair: damaged mtDNA fails to obtain functional copies via fusion, causing progressive mutation-load accumulation and aggravating OXPHOS defects and ROS production (Vidoni et al., 2013). Excessive fission causes mitochondrial fragmentation, which prevents mitochondria from achieving effective fusion and functional coordination. Meanwhile, fragmented mitochondria are more vulnerable to ROS-induced damage, and their OXPHOS function is significantly impaired with a further increase in electron

leakage rate, forming a positive feedback loop of dynamics imbalance and increased ROS production (Adebayo et al., 2021). In addition, dynamics imbalance impairs the efficiency of mitophagy, resulting in the failure to timely clear damaged mitochondria that continuously generate ROS, thus further exacerbating oxidative stress and mtDNA damage (Chen et al., 2023).

This cascade amplification effect has been fully verified in AD model animals and patients. Studies have shown that in AD model mice, the activity of OXPHOS Complex I is reduced and ROS production is increased, which subsequently leads to the downregulation of MFN2 expression and enhanced DRP1 Ser616 phosphorylation, resulting in mitochondrial fragmentation. Fragmented mitochondria further exacerbate mtDNA mutation load and ROS generation, ultimately causing neuronal damage and cognitive decline (Chen et al., 2023; Swerdlow et al., 2014). Additionally, in the brains of AD patients, regions with a higher mtDNA mutation load exhibit significantly elevated levels of oxidative stress markers and a higher degree of mitochondrial fragmentation, with a positive correlation observed among the three indices, which indicates the existence of this cascade amplification effect (Dhapola et al., 2024).

3.2. Mitophagy dysfunction & calcium homeostasis imbalance in synergistic apoptosis activation

A close mutual regulatory relationship exists between mitophagy dysfunction and calcium homeostasis imbalance, and the two synergistically activate the intrinsic apoptotic pathway, serving as a pivotal

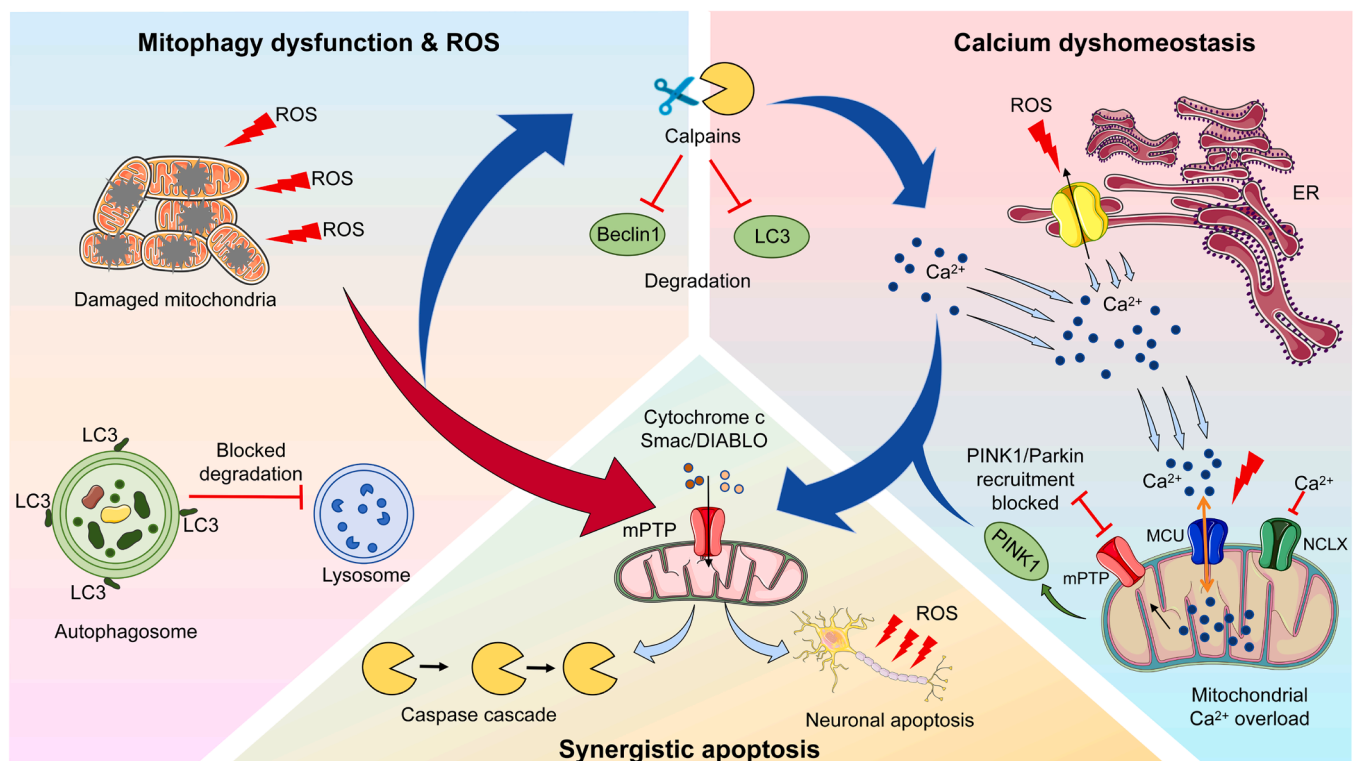


Fig. 4. Synergistic induction of neuronal apoptosis by mitophagy dysfunction, ROS overproduction, and calcium dyshomeostasis. Mitophagy dysfunction, ROS overproduction, and Ca^{2+} dyshomeostasis act synergistically to drive neuronal apoptosis. Damaged mitochondria generate excessive ROS, which activate calpains to degrade Beclin1 and LC3, thereby blocking autophagosome-lysosome fusion and impairing the clearance of damaged mitochondria. This defective mitophagy leads to the accumulation of ROS-producing mitochondria, creating a self-reinforcing cycle. Concurrently, ROS disrupts ER calcium homeostasis, promoting Ca^{2+} release, while also enhancing MCU activity and inhibiting the sodium-calcium exchanger (NCLX), resulting in mitochondrial Ca^{2+} overload. This calcium overload further blocks the recruitment of PINK1 and Parkin, exacerbating mitophagy dysfunction. Together, ROS overproduction and calcium dyshomeostasis trigger the opening of the mPTP, leading to the release of pro-apoptotic factors such as cytochrome c and Smac/DIABLO. These factors activate the caspase cascade, ultimately driving neuronal apoptosis. ROS (reactive oxygen species), Ca^{2+} (calcium), ER (endoplasmic reticulum), MCU (mitochondrial calcium uniporter), NCLX (sodium-calcium exchanger), PINK1 (PTEN-induced kinase 1), mPTP (mitochondrial permeability transition pore), Smac/DIABLO (second mitochondria-derived activator of caspases/direct IAP-binding protein with low pI). Figure created with PowerPoint.

mechanism underlying neuronal loss (Fig. 4).

Mitophagy dysfunction disrupts calcium homeostasis through multiple pathways. First, mitophagy dysfunction leads to the accumulation of damaged mitochondria, which exhibit OXPHOS dysfunction and increased ROS production. Excessive ROS can oxidatively modify mitochondrial calcium regulatory proteins (e.g., MCU, NCLX), enhancing MCU activity and inhibiting NCLX function. This results in increased mitochondrial Ca^{2+} uptake and decreased release, triggering mitochondrial Ca^{2+} accumulation (Chang and Zou, 2025; Wu et al., 2019). Second, mitophagy dysfunction impairs the endoplasmic reticulum (ER) calcium storage and release function: autophagy regulates ER calcium homeostasis by degrading ER-associated proteins, and impaired autophagy causes an increase in ER calcium storage. Meanwhile, ROS damages ER Ca^{2+} channels (e.g., IP3R), promoting ER Ca^{2+} release and leading to elevated cytoplasmic Ca^{2+} concentration, which in turn facilitates mitochondrial Ca^{2+} uptake (Toglia et al., 2016). In addition, energy metabolism failure induced by mitophagy dysfunction impairs the function of plasma membrane Ca^{2+} -ATPase and ER Ca^{2+} -ATPase, inhibiting cytoplasmic Ca^{2+} clearance and further exacerbating calcium homeostasis imbalance (Joseph et al., 2019). Conversely, calcium homeostasis imbalance significantly suppresses mitophagic activity. Excessive mitochondrial Ca^{2+} accumulation can induce the opening of the mPTP, leading to a decrease in mitochondrial membrane potential - an essential prerequisite for the stabilization and activation of PINK1. The loss of membrane potential inhibits the accumulation of PINK1 on the surface of damaged mitochondria, thereby blocking the PINK1-Parkin-mediated mitophagic pathway (Barazzuol et al., 2020). At the same time, elevated cytoplasmic Ca^{2+} concentration activates calcium-dependent proteases (e.g., calpains), which degrade autophagy-related proteins (e.g., Beclin1, LC3) and inhibit the formation and maturation of autophagosomes (Wang et al., 2025). Furthermore, oxidative stress triggered by calcium homeostasis imbalance impairs lysosomal function, inhibits the fusion of autophagosomes with lysosomes, and further hinders the degradation process of mitophagy (Wu et al., 2022).

Mitophagy dysfunction and calcium-homeostasis imbalance synergistically activate the intrinsic apoptotic pathway. On the one hand, accumulation of damaged mitochondria from defective mitophagy, combined with mitochondrial Ca^{2+} overload driven by disrupted calcium homeostasis, cooperatively facilitates mPTP opening. Damaged mitochondria display lowered membrane potential and heightened Ca^{2+} sensitivity; excess Ca^{2+} further induces mPTP opening, triggering membrane-potential collapse, outer-membrane rupture, and release of pro-apoptotic factors such as cytochrome c and Smac/DIABLO (Wang and Zheng, 2019). On the other hand, mitophagy dysfunction prevents the timely clearance of pro-apoptotic factors, while calpains activated by calcium homeostasis imbalance further promote the release of these factors and simultaneously trigger the caspase cascade reaction, ultimately resulting in neuronal apoptosis (Tsapras and Nezis, 2017). In addition, oxidative stress and energy metabolism failure induced by the synergy of the two further activate the apoptotic pathway and exacerbate neuronal loss through multiple approaches including DNA damage and cytoskeletal disruption (Torii et al., 2023).

In AD model animals, ameliorating mitophagy via pharmacological or genetic approaches (e.g., Parkin overexpression) can significantly reduce mitochondrial Ca^{2+} levels, restore calcium homeostasis, inhibit the activation of the apoptotic pathway, and decrease neuronal apoptosis (Yu et al., 2024). Moreover, restoring calcium homeostasis by inhibiting MCU activity can also markedly enhance mitophagic activity, clear damaged mitochondria, and improve neuronal survival (Cai et al., 2022). Furthermore, the concept of synergistic regulation is vividly illustrated by the converging toxic effects of A β and hyperphosphorylated Tau on mitochondrial dynamics and synaptic energy supply. Rather than acting in isolation, these two pathological hallmarks synergistically dismantle the mitochondrial network. Specifically, A β oligomers abnormally interact with the mitochondrial fission protein

Drp1, inducing excessive mitochondrial fragmentation (Manczak et al., 2011). Concurrently, hyperphosphorylated Tau aggregates disrupt the microtubule tracks and impair motor proteins (kinesin and dynein), physically halting the axonal transport of these mitochondria. In a classic *in vivo* model, this synergistic attack—where A β fragments the mitochondria and Tau prevents their transport—results in a severe depletion of healthy mitochondria at the presynaptic terminals. This dual-hit mechanism perfectly exemplifies how synergistic regulation leads to profound synaptic energy starvation and accelerated cognitive decline, far exceeding the toxicity of either protein alone (Zhang et al., 2021a). This mutual regulatory relationship suggests that targeting the synergistic regulatory pathways of mitophagy and calcium homeostasis may represent a more effective intervention strategy for AD.

3.3. Mitochondrial dysfunction-aging microenvironment crosstalk in AD

Mitochondrial dysfunction is not only a core intracellular pathological change in AD but also exhibits a close bidirectional regulatory relationship with the aging brain microenvironment of AD, which mainly includes neuroinflammation and metabolic disorders. The two interact and promote each other, collectively driving the pathological progression of AD (Fig. 5).

The Mitochondrial Cascade Hypothesis for AD proposed by Swerdlow and colleagues states that an individual's genetic background (determined by mtDNA haplogroups) and environmental factors jointly establish the basal functional level and aging rate of mitochondria (Swerdlow et al., 2010). With aging, the efficiency of the mitochondrial electron transport chain declines gradually, which directly triggers the explosive production of ROS. Excessive ROS not only damages mtDNA but also inhibits the activity of key metabolic enzymes such as α -ketoglutarate dehydrogenase, ultimately leading to the occurrence of mitochondrial dysfunction (Tretter and Adam-Vizi, 2005). Notably, recent studies have further revealed a novel role of mitochondria as a hub for immune regulation, providing an important supplement to understanding their role in AD pathogenesis. When mitochondria are damaged, the mtDNA they release can escape into the cytoplasm as damage-associated molecular patterns (DAMPs), triggering microglia-mediated sterile neuroinflammation by activating the cGAS-STING signaling pathway—a process that directly exacerbates AD-related neurodegeneration (Tian et al., 2025). Moreover, a landmark study demonstrated that activated microglia release fragmented, damaged mitochondria which subsequently induce the conversion of naive astrocytes into the neurotoxic A1 phenotype (Joshi et al., 2019). This mitochondria-mediated microenvironment crosstalk propagates an inflammatory cascade from microglia to astrocytes, and ultimately back to neurons, driving widespread neuronal death and highlighting mitochondria as central signaling hubs in AD neuroinflammation.

In fact, mitochondrial dysfunction itself is a key inducer of neuroinflammation. In addition to mtDNA, other DAMP molecules released by damaged mitochondria, such as mitochondrial-derived peptides and ROS, can also activate neuroinflammatory pathways through different mechanisms. Specifically, mtDNA released into the cytoplasm can bind to pattern recognition receptors such as TLR9 or activate microglia and astrocytes via the cGAS-STING pathway, promoting the release of inflammatory cytokines, chemokines and nitric oxide (NO) (Saber et al., 2022). ROS can directly activate the NF- κ B signaling pathway to drive the transcription and expression of inflammatory cytokines, and can also further enhance their transcriptional activity by oxidatively modifying the promoter regions of inflammatory cytokine genes (Shi et al., 2016). Furthermore, energy metabolism failure caused by mitochondrial dysfunction impairs the phagocytic function and polarization state of microglia, promoting their transformation into the pro-inflammatory phenotype, thereby further amplifying the neuroinflammatory response (Ren et al., 2020). Interestingly, neuroinflammation is not merely a passive outcome; it can in turn exacerbate mitochondrial dysfunction through multiple pathways, forming a positive feedback

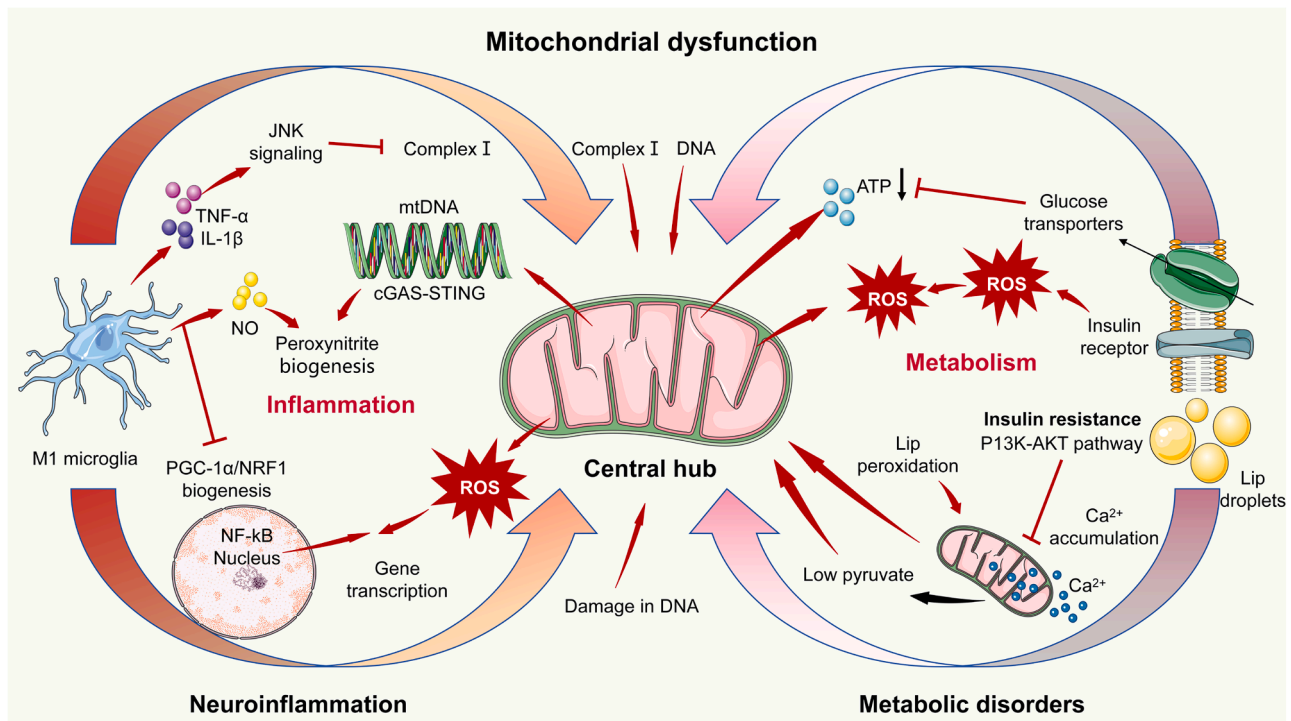


Fig. 5. Mitochondrial dysfunction as a central hub linking neuroinflammation and metabolic disorders. Mitochondrial dysfunction acts as a central hub that reciprocally links neuroinflammation and metabolic disorders. In the neuroinflammatory axis, pro-inflammatory microglia release pro-inflammatory cytokines including TNF- α and IL-1 β , which activate JNK signaling to suppress mitochondrial Complex I activity. Concurrently, NO promotes peroxynitrite biogenesis, inhibiting peroxisome PGC-1 α and NRF1-mediated mitochondrial biogenesis. Mitochondrial DNA (mtDNA) released from damaged mitochondria activates the cGAS-STING pathway, while excessive ROS generated by dysfunctional mitochondria activates NF- κ B in the nucleus, driving pro-inflammatory gene transcription and amplifying the inflammatory response. In the metabolic axis, mitochondrial dysfunction reduces ATP production, and ROS impairs glucose transporters and insulin receptors, leading to insulin resistance and dysregulation of the PI3K-AKT signaling pathway. Additionally, lipid peroxidation, intracellular Ca²⁺ accumulation, and reduced pyruvate levels further exacerbate mitochondrial damage. These interconnected pathways form a vicious cycle, where neuroinflammation and metabolic disorders mutually reinforce mitochondrial dysfunction, contributing to the progression of neurodegenerative and metabolic pathologies. TNF- α (tumor necrosis factor- α), IL-1 β (interleukin-1 β), JNK (c-Jun N-terminal kinase), mtDNA (mitochondrial DNA), cGAS-STING (cyclic GMP-AMP synthase-stimulator of interferon genes), NO (nitric oxide), PGC-1 α (peroxisome proliferator-activated receptor gamma coactivator 1-alpha), NRF1 (nuclear respiratory factor 1), NF- κ B (nuclear factor kappa B), ROS (reactive oxygen species), ATP (adenosine triphosphate), PI3K-AKT (phosphatidylinositol 3-kinase-protein kinase B), Ca²⁺ (calcium). Figure created with PowerPoint.

loop of inflammation-mitochondrial damage. On the one hand, inflammatory cytokines released by microglia and astrocytes can act directly on neuronal mitochondria: for example, TNF- α can phosphorylate the subunits of OXPHOS Complex I by activating the JNK signaling pathway, reducing its activity; IL-1 β can upregulate the expression of iNOS to promote NO production, and NO can react with ROS to form peroxynitrite, which causes significant damage to mitochondrial function (Higuchi et al., 1998; Win et al., 2014). On the other hand, inflammatory cytokines can inhibit the expression of genes related to mitochondrial biogenesis such as PGC-1 α and NRF1, reducing mitochondrial quantity and further lowering the efficiency of energy metabolism (Jung et al., 2021). In addition, blood-brain barrier damage induced by neuroinflammation leads to the infiltration of peripheral immune cells into the central nervous system and the release of more inflammatory cytokines. This process not only exacerbates the local inflammatory response but also further strengthens the positive feedback loop of inflammation-mitochondrial damage, ultimately accelerating the pathological progression of AD (Rivera-Escalera et al., 2014).

A bidirectional regulatory relationship also exists between mitochondrial dysfunction and metabolic disorders. Obvious metabolic disorders are present in the AD brain, mainly characterized by reduced glucose metabolism, insulin resistance, abnormal lipid metabolism and other abnormalities, and mitochondrial dysfunction is one of the core causes of these metabolic disorders (Singulani et al., 2021). Mitochondrial OXPHOS dysfunction leads to insufficient ATP production, which can inhibit the expression and activity of glucose transporters, reduce

the brain's uptake and utilization of glucose, and result in decreased glucose metabolism (Fernandez-Vizarra and Zeviani, 2021). Meanwhile, mitochondrial dysfunction can interfere with the insulin signaling pathway, leading to insulin resistance. The PI3K-AKT signaling pathway downstream of the insulin receptor not only participates in the regulation of glucose metabolism but also affects mitochondrial function by regulating mitochondrial biogenesis and mitophagy. Excessive mitochondrial ROS production can oxidatively modify the insulin receptor and downstream signaling molecules, inhibit the activity of this pathway, and induce insulin resistance (Chen and Zhong, 2013). In addition, mitochondrial dysfunction can impair the activity of lipid metabolism-related enzymes (e.g., fatty acid synthase, fatty acid oxidase), leading to lipid accumulation and oxidation, which further aggravates mitochondrial damage (Mi et al., 2023). Conversely, metabolic disorders can exacerbate mitochondrial dysfunction through multiple pathways. Reduced glucose metabolism leads to insufficient energy supply to neurons, further inhibiting mitochondrial biogenesis and repair and exacerbating OXPHOS dysfunction (Baloyannis, 2006; Nam et al., 2023). At the same time, a decrease in pyruvate, an intermediate product of glucose metabolism, can inhibit the tricarboxylic acid cycle and further reduce ATP production (Butterfield and Halliwell, 2019). Insulin resistance can reduce mitophagic activity by inhibiting the PI3K-AKT signaling pathway, leading to the accumulation of damaged mitochondria; additionally, insulin resistance can disrupt mitochondrial calcium homeostasis, promote mitochondrial Ca²⁺ accumulation, and exacerbate mitochondrial damage (Hees et al., 2024). Furthermore,

lipid accumulation and oxidation caused by abnormal lipid metabolism can generate a large number of lipid peroxidation products, which can directly damage mitochondrial membrane lipids and proteins, disrupt mitochondrial structure and function, and promote ROS production (Pessayre, 2007).

In AD-patient brains, the severity of mitochondrial dysfunction, neuroinflammation and metabolic disturbances positively correlate with one another and closely relate to cognitive impairment. Regions exhibiting cerebral glucose hypometabolism in AD frequently show concurrent mitochondrial OXPHOS defects, microglial activation and increased pro-inflammatory cytokines, pointing to synergistic interactions among the three components (Tondo et al., 2021). This bidirectional regulatory relationship indicates that the pathological progression of AD is the result of the combined action of intracellular mitochondrial dysfunction and extracellular microenvironmental disorders. Targeting multiple links of this regulatory network may be the key to improving the efficacy of intervention strategies for AD.

3.4. Interplay between mitochondrial transport defects and other mechanisms

Mitochondrial transport defects do not exist in isolation but exhibit close crosstalk with mtDNA mutations, oxidative stress, mitochondrial dynamics imbalance, mitophagy dysfunction, calcium homeostasis disorder and other mechanisms, forming a complex regulatory network that collectively exacerbates synaptic damage and neuronal loss (Sterky et al., 2011).

MtDNA mutations and oxidative stress act as key upstream factors leading to mitochondrial transport defects. OXPHOS dysfunction caused by mtDNA mutations impairs mitochondrial energy supply, and mitochondrial transport itself is an ATP-dependent process; insufficient energy supply directly reduces the speed and efficiency of mitochondrial transport (Sterky et al., 2011). Meanwhile, excessive ROS production induced by mtDNA mutations can oxidatively modify mitochondrial transport-related proteins, resulting in their conformational changes, reduced activity and even degradation (Wu et al., 2023). Studies have confirmed that ROS can oxidize the iron-sulfur cluster structure of Miro1, disrupting its binding to kinesins and thus inhibiting anterograde transport. Additionally, ROS can oxidize the motor domain of KIF5A, reducing its ATPase activity and impairing transport motility (Alshaabi et al., 2021; Li et al., 2025b). Furthermore, ROS can damage microtubule structures, disrupt the mitochondrial transport track and further aggravate transport defects. Mitochondrial transport defects are not an isolated pathological event, but also maintain tight mutual regulation with mitochondrial dynamics imbalance; the two can form a vicious cycle that continuously exacerbates mitochondrial dysfunction (Goldblum et al., 2021). Mitochondrial morphology is closely correlated with transport efficiency: fused tubular mitochondria bind more readily to kinesins for long-distance transport, whereas fragmented mitochondria show a significant decline in transport efficiency (Li et al., 2025a). Mitochondrial fragmentation caused by dynamics imbalance in AD directly impairs the binding capacity of mitochondria to kinesins and inhibits anterograde transport. At the same time, fragmented mitochondria are more prone to aggregate in axons, forming transport blockages that further hinder the transport of other mitochondria (Correia et al., 2016). Conversely, insufficient mitochondrial number at synaptic sites due to transport defects prevents mitochondria from achieving functional complementation with other mitochondria through fusion, and damaged mitochondria cannot be timely transported back to the soma for repair, which further exacerbates dynamics imbalance and mitochondrial fragmentation (Tsousis et al., 2011).

In addition, mitochondrial transport defects and mitophagy dysfunction exhibit a marked mutual promotion effect, which together drive the abnormal accumulation of damaged mitochondria in cells. A core function of mitochondrial retrograde transport is to deliver senescent/damaged mitochondria from synaptic sites back to the soma for

mitophagic degradation. However, retrograde transport defects in AD prevent damaged mitochondria from reaching the soma in a timely manner, thus inhibiting the initiation of mitophagy (Han et al., 2020). Meanwhile, mitophagy dysfunction results in the ineffective degradation of damaged mitochondria in the soma; these mitochondria may re-enter the transport system or interfere with the transport of normal mitochondria, further exacerbating transport defects (Tammineni et al., 2017). Furthermore, oxidative stress and energy metabolism failure induced by mitophagy dysfunction can further inhibit mitochondrial transport by damaging transport-related proteins and microtubule structures (López-Doménech and Kittler, 2023). Calcium homeostasis disorder is also an important regulatory factor for mitochondrial transport defects: it can directly exacerbate abnormal mitochondrial transport through multiple pathways, and the two can form a vicious cycle that continuously amplifies pathological effects. Elevated cytoplasmic Ca^{2+} concentration can bind to Miro1, promoting its ubiquitin-dependent degradation, disrupting the binding of mitochondria to kinesins and inhibiting anterograde transport (Endoni et al., 2025). Meanwhile, Ca^{2+} can activate calpains to degrade microtubule-associated proteins, disrupting microtubule stability and impairing the mitochondrial transport track (Hoskison et al., 2007). In addition, mitochondrial swelling and decreased membrane potential caused by excessive mitochondrial Ca^{2+} accumulation can impair mitochondrial morphology and function, reducing its transport efficiency (Prentice et al., 2015). Conversely, insufficient mitochondrial number at synaptic sites due to transport defects impairs the effective uptake of cytoplasmic Ca^{2+} by mitochondria, further exacerbating calcium homeostasis disorder and forming a vicious cycle (Su et al., 2014).

Mitochondrial transport defects do not merely form local vicious cycles with a single mitochondrial dysfunction mechanism; as a core hub of mitochondrial dysfunction, they also systematically exacerbate abnormalities in other mitochondria-related mechanisms and exert cross-regulation with the core pathological changes of AD, collectively driving the overall pathological progression of AD (Area-Gomez and Schon, 2017a). Energy supply shortage and oxidative stress caused by insufficient mitochondrial number at synaptic sites can further promote mtDNA mutations, dynamics imbalance, mitophagy dysfunction and calcium homeostasis disorder (Wei et al., 2025). Meanwhile, synaptic dysfunction induced by transport defects impairs neural signal transmission, which in turn affects neuronal physiological functions and promotes A β production and tau hyperphosphorylation. In turn, abnormalities in A β and tau protein can exacerbate mitochondrial transport defects through multiple pathways, forming a complex regulatory network of “transport defects - abnormalities in other mechanisms - AD pathogenesis” (Wang et al., 2019a). Studies have found ameliorating mitochondrial transport by overexpressing KIF5A or Miro1 can significantly increase the number of mitochondria at synaptic sites and restore mitochondrial function, while simultaneously reducing ROS levels, improving mitochondrial dynamics balance, enhancing mitophagic activity and restoring calcium homeostasis. Ultimately, this leads to a reduction in A β deposition and tau aggregation, and an improvement in cognitive function (Jeong et al., 2022; Wang et al., 2019a). These findings suggest that mitochondrial transport defects are a key link connecting multiple mitochondrial dysfunction mechanisms, and targeting mitochondrial transport may simultaneously ameliorate multiple pathological processes, holding important therapeutic potential for AD treatment.

3.5. The regulatory role of mitochondrial heterogeneity in the pathological progression of AD

Mitochondria are not a homogeneous population of organelles, but exhibit significant functional and structural heterogeneity across different cell types, brain regions, and even subcellular compartments within the same cell. This heterogeneity plays a crucial yet underappreciated regulatory role in the pathological progression of AD (Kim and

Mook-Jung, 2019; Liao et al., 2024).

From the perspective of cell type specificity, mitochondria in different brain cell types possess distinct metabolic characteristics and functional requirements. As highly polarized, energy-dependent cells, neuronal mitochondria are predominantly localized in energy-demanding regions such as synapses and axons. Their dysfunction directly causes synaptic transmission deficits and neuronal death, with pyramidal neurons in the hippocampal CA1 region and cerebral cortex being the most sensitive to mitochondrial damage, which is closely related to their high metabolic rate and long axonal structure (Cicali et al., 2026). The mitochondrial functional status of microglia directly determines their polarization phenotype. Resting microglia primarily rely on OXPHOS for energy supply. However, upon exposure to A β , microglial mitochondria undergo a rapid metabolic reprogramming characterized by a shift from OXPHOS to glycolysis. This metabolic shift, accompanied by mitochondrial fragmentation and excessive ROS production, drives microglia into a pro-inflammatory state, releasing neurotoxic cytokines while significantly impairing their phagocytic capacity for A β clearance (Baik et al., 2019; Ulland et al., 2017). Conversely, astrocytes are highly glycolytic under physiological conditions but strictly rely on robust mitochondrial function for lipid metabolism, glutamate clearance, and antioxidant production. In the AD microenvironment, astrocytic mitochondrial dysfunction impairs their ability to buffer synaptic glutamate, leading to neuronal excitotoxicity, and reduces the provision of antioxidant support (such as glutathione) and metabolic substrates (like lactate) to neurons, thereby exacerbating neuronal vulnerability (Kim et al., 2024).

Furthermore, mitochondrial heterogeneity is also reflected at the macroscopic level across different brain regions. Compared to the vulnerable hippocampus and cortex, mitochondria in regions like the visual cortex and cerebellum remain relatively intact in the early stages of AD, which explains why neuronal loss in these regions only occurs in the late stages of the disease (Baranich et al., 2026; Yang et al., 2024). In addition, mitochondria within the same neuron also exhibit significant subcellular heterogeneity. Synaptic mitochondria and somatic mitochondria show distinct differences in morphology, function, and proteomic profiles. The former have higher respiratory activity and calcium buffering capacity but are also more vulnerable to the toxic damage caused by A β and tau proteins. This selective damage to synaptic mitochondria is one of the core mechanisms underlying early synaptic dysfunction and cognitive decline in AD (Cicali et al., 2026; Pekkurnaz and Wang, 2022).

4. Mitochondria-targeted intervention strategies for AD

4.1 Mitochondria-targeted antioxidants

The Mitochondrial Cascade Hypothesis for AD clearly identifies oxidative stress as a core driver of disease initiation and progression (Swerdlow et al., 2010). As the primary site of cellular oxidative metabolism and the major source of ROS production, mitochondria form a vicious cycle between functional disturbance and oxidative damage that plays a pivotal role in AD pathological progression. Excessive ROS accumulation triggers lipid peroxidation, protein oxidation and nucleic acid damage, which in turn cause mitochondrial structural disruption and functional decline, ultimately inducing neuronal apoptosis and cognitive impairment. Targeted regulation of mitochondrial oxidative stress has thus emerged as a crucial therapeutic strategy for AD (Maruszak and Żekanowski, 2011). Although traditional antioxidants such as vitamin E, vitamin C and coenzyme Q10 (CoQ10) exhibit definite antioxidant activity in *in vitro* experiments, they have failed to yield satisfactory efficacy in AD clinical trials (Dysken et al., 2014; Galasko et al., 2012). The core reason for this predicament is the lack of mitochondrial targeting ability in traditional antioxidants: these molecules cannot effectively penetrate the double mitochondrial membrane and are unable to reach therapeutically effective concentrations within the mitochondrial matrix. This prevents their precise intervention in the

production and accumulation of mitochondria-derived ROS, ultimately limiting their clinical application value.

To overcome the pharmacokinetic limitations of traditional drugs, mitochondria-targeted antioxidants (MTAs) have been developed (Reddy and Reddy, 2017; Şerban et al., 2025). Such drugs couple antioxidant active moieties with targeting moieties via chemical modification, with the most commonly used targeting strategies involving the introduction of lipophilic cations (e.g., triphenylphosphonium, TPP⁺) or specific targeting peptides. Leveraging the electrochemical gradient of mitochondrial membrane potential ($\Delta\Psi_m$) or ligand-receptor specific binding, these drugs accumulate in mitochondria to levels hundreds of times higher than in the cytoplasm, enabling precision scavenging of ROS and targeted protection of mitochondrial function, and thus providing a novel approach for AD treatment (Murphy and Smith, 2007). Among representative MTAs, TPP⁺-conjugated agents occupy an important position. Mitochondrial coenzyme Q10 (MitoQ) is synthesized by linking the active component of CoQ10 (ubiquinone) to the TPP⁺ cation via an aliphatic carbon chain. Through the lipophilic and positively charged properties of TPP⁺, MitoQ accumulates in the mitochondrial matrix at concentrations 100–500 times higher than in the cytoplasm (Manczak et al., 2010). In APP/PS1 transgenic mouse models, MitoQ treatment significantly reduces free A β levels in the cerebral cortex and hippocampus, decreases the accumulation of oxidative stress markers such as lipid peroxides, and ameliorates synaptic loss and spatial memory deficits. Its mechanism of action not only involves the direct scavenging of peroxyl radicals, but also the promotion of mitochondrial biogenesis by upregulating peroxisome proliferator-activated receptor γ coactivator 1 α (PGC-1 α), achieving dual protection of mitochondrial function (Oliver and Reddy, 2019b).

Another typical TPP⁺-conjugated MTA is Mito-Tempo, formed by the conjugation of piperidine nitroxide (Tempo) with TPP⁺. As a specific mimetic of superoxide dismutase (SOD) activity, it precisely scavenges mitochondrial superoxide, effectively inhibits A β -induced mitochondrial ROS production and neuronal apoptosis, and maintains the stabilization of mitochondrial membrane potential in AD models (Du et al., 2022). MitoVitE (Mito-Vitamin E), another MTA based on the TPP⁺ targeting strategy, is prepared by chemically conjugating α -tocopherol with TPP⁺, which elevates its antioxidant efficiency in mitochondria to more than 350 times that of ordinary vitamin E. It can effectively block A β -induced mitochondrial lipid peroxidation and inhibit oxidative stress-mediated neuronal necrosis (Manabe et al., 2019). In addition, SkQ1 represents a unique member of TPP⁺-conjugated MTAs, linking plant-derived plastoquinone to the targeting group via decyldiphenylphosphonium (Stefanova et al., 2019). It exhibits extremely strong antioxidant activity at low effective concentrations. SkQ1 prevents A β -induced long-term potentiation (LTP) impairment and improves cognitive function in AD rats. It also possesses a unique anti-aging property, significantly extending the lifespan of premature aging mice and reversing hippocampal pathological changes (Stefanova et al., 2016).

Beyond TPP⁺-mediated MTAs, agents based on non-TPP⁺ targeting strategies have also demonstrated remarkable advantages, with the small-molecule tetrapeptide SS-31 from the Szeto-Schiller peptide family being the most representative (Chaturvedi and Beal, 2008). SS-31 achieves inner mitochondrial membrane targeting through specific binding to cardiolipin, a mitochondria-specific phospholipid. Unlike TPP⁺-based drugs that rely on mitochondrial membrane potential, SS-31 stabilizes mitochondrial cristae structure, protects electron flux in the electron transport chain (especially Complexes I and II), and reduces ROS production caused by electron leakage. It is also capable of crossing the blood-brain barrier (BBB) (Ding et al., 2021). In the brain of 5 $\times$ FAD mice, the linker between cyclosporine A (CsA) and SS-31 is cleaved under the high ROS microenvironment within the damaged mitochondria of microglia and neurons, and CsA and SS-31 are released while scavenging ROS in the mitochondrial compartment. The liberated CsA and SS-31 act synergistically to restore mitochondrial membrane

potential and the balance between mitochondrial fission and fusion, thereby protecting neurons from apoptosis, reducing microglial activation in the brain, and ameliorating neuroinflammation and cognitive impairment in 5 × FAD mice (Qian et al., 2025). Natural astaxanthin, with its unique mitochondrial membrane enrichment property, serves as an important supplement to non-TPP* targeted MTAs. Its special molecular structure allows it to cross the cell membrane and insert into the mitochondrial membrane, forming a natural antioxidant barrier (Villalain, 2025). In AD models, astaxanthin not only directly scavenges ROS, but also promotes neurogenesis and improves synaptic plasticity by regulating the Erk1/2 and CREB signaling pathways (Oliyaee et al., 2023).

In summary, via precise mitochondrial-targeting design, MTAs circumvent pharmacokinetic drawbacks of conventional antioxidants, achieving targeted ROS scavenging and mitochondrial protection — a key advance in AD antioxidant therapy. Diverse MTAs with distinct targeting modes and functional properties exert favorable neuro-protective and cognition-improving effects in AD pre-clinical models (Table 1). Notably, the advanced clinical progress of SS-31 raises translational prospects for MTA-based AD therapeutics (Ding et al., 2021). Future work should further optimize MTAs for brain delivery, target specificity and bioavailability. In addition, combinatorial benefits of MTAs with mitophagy modulators, mitochondrial-dynamics regulators, mtDNA-protecting agents and other strategies should be explored. Multi-target, multi-pathway combined modulation may comprehensively ameliorate mitochondrial pathological abnormalities and foster clinical advances in precision AD therapy.

4.1. Mitochondrial mtDNA repair and protective agents

The mtDNA mutation load is markedly elevated in the brains of AD patients, particularly in genes encoding key subunits of the electron transport chain, which are highly susceptible to damage. This pathological change directly triggers a cellular energy metabolism crisis and forms a vicious cycle with pathological processes such as oxidative stress. Unlike the nucleus, mitochondria lack a sophisticated DNA repair system; their damage repair mainly relies on the base excision repair (BER) pathway with limited repair capacity (Zhu et al., 2004). Therefore, enhancing mtDNA repair capacity or protecting mtDNA from damage has become an important strategy for AD intervention.

Current drug research and development in this field mainly focuses on two directions: activating upstream signaling pathways of mitochondrial biogenesis (e.g., the SIRT1-PGC1 α -TFAM axis) to indirectly increase the copy number of healthy mtDNA, and directly enhancing the activity of DNA repair enzymes (Santos et al., 2013). Among them, NAD⁺ precursors nicotinamide riboside (NR) and nicotinamide mononucleotide (NMN) are important research hotspots. The decline in brain NAD⁺ levels in AD patients leads to reduced activity of the deacetylase SIRT1. Supplementation with NR or NMN can effectively activate SIRT1, which in turn activates PGC-1 α , the master regulator of mitochondrial

biogenesis, through deacetylation. PGC-1 α upregulates the expression of mitochondrial transcription factor A (TFAM), which not only regulates mtDNA transcription but also wraps and stabilizes mtDNA like histones, protecting it from ROS-induced damage (Lagouge et al., 2006; Lautrup et al., 2019; Mouchiroud et al., 2013). A study confirmed that NR treatment in AD mice significantly reduces the DNA damage response and ameliorates mitochondrial function and cognitive behaviors (Hou et al., 2018). In addition, epigenetic modulators such as histone deacetylase inhibitors (HDACi), including Vorinostat and Valproic acid, have also shown potential. These drugs are found to upregulate the expression of key enzymes involved in mtDNA repair (e.g., OGG1 and APE1). In AD models, they effectively reduce the levels of mtDNA oxidative damage markers by enhancing the efficiency of the BER pathway (Ateya et al., 2025; De Simone and Milelli, 2019). Latrepirdine (Dimebon) represents another class of drugs with a unique mechanism. Although its results in phase III AD clinical trials are inconsistent, mechanistic studies have revealed its important mitochondrial protective effects. This drug can block A β -induced opening of the mPTP, protect cells from A β toxicity at nanomolar concentrations, and more importantly, improve mitochondrial membrane fluidity to indirectly maintain the structural stability of the mitochondrial membrane to which mtDNA attaches, thereby reducing mutation accumulation (Zhang et al., 2010). Furthermore, berberine, a natural small-molecule compound, also shows promising prospects. As an isoquinoline alkaloid that can cross the blood-brain barrier, it upregulates the expression of SIRT1 and PGC-1 α by activating the AMPK signaling pathway, significantly increasing mtDNA copy number and ATP levels in hippocampal neurons of AD mice, and exhibiting definite mtDNA protective effects and the ability to promote mitochondrial biogenesis (Akbar et al., 2021; Cai et al., 2016; Singh et al., 2019). Among intervention strategies targeting apolipoprotein E4 (ApoE4), the greatest genetic risk factor for AD, ApoE-mimetic peptides (e.g., COG1410) have distinctive features. ApoE4 fragments not only aggregate extracellularly but also translocate into neuronal mitochondria and bind to damaged mtDNA. COG1410 can reduce mtDNA damage and cytotoxicity by competitively inhibiting the interaction between ApoE4 and mitochondria (Qiao et al., 2024; Zhang et al., 2021b). Meanwhile, the research and development of drugs directly targeting repair enzymes is also advancing. OGG1, a key enzyme responsible for excising oxidatively damaged bases in mtDNA, has small-molecule activators entering the early research and development stage, which aim to directly enhance the rate of DNA repair in mitochondria and prevent respiratory chain dysfunction caused by mutation accumulation (Oka et al., 2021).

In summary, As central contributors to energy metabolic disorders in AD, mtDNA damage and defective repair represent promising intervention targets. Pre-clinical studies have identified multiple agents with mtDNA-protective and mitochondria-improving effects via diverse strategies: activating the SIRT1-PGC1 α -TFAM axis, boosting the BER pathway, preserving mitochondrial membrane integrity, targeting ApoE4-driven pathogenic mechanisms, and directly activating repair

Table 1
Mitochondria-targeted therapeutic drugs for AD – Mitochondria-targeted antioxidants.

Representative drugs	Clinical status	Target / Core mechanism	Main pharmacological effects	References
MitoQ	Phase II clinical trial	Targets mitochondria via TPP*, scavenges ROS, upregulates PGC-1 α .	Reduces A β deposition and oxidative stress, improves synaptic function and memory.	(Oliver and Reddy, 2019b)
Mito-Tempo	Preclinical	TPP*-conjugated SOD mimetic for mitochondria.	Scavenges mitochondrial superoxide anion, inhibits neuronal apoptosis, stabilizes membrane potential.	(Du et al., 2022)
MitoVitE	Preclinical	TPP*-targeted high-potency vitamin E.	Blocks mitochondrial lipid peroxidation, inhibits oxidative stress-mediated neuronal necrosis.	(Manabe et al., 2019)
SkQ1	Phase III failed	Targets mitochondria, potent antioxidant and anti-aging agent.	Improves LTP impairment and cognitive function, reverses hippocampal pathological changes.	(Stefanova et al., 2016)
SS-31	Phase III clinical trial ongoing	Binds cardiolipin, targets inner mitochondrial membrane, stabilizes cristae structure.	Crosses blood-brain barrier, protects electron transport chain, alleviates neuroinflammation.	(Ding et al., 2021)
Astaxanthin	Phase II clinical trial	Enriches in mitochondrial membrane, directly scavenges ROS.	Regulates signaling pathways, promotes neurogenesis and synaptic plasticity.	(Oliyaee et al., 2023)

enzymes (Table 2). These findings deepen our understanding of AD pathogenesis and inform the development of precision-targeted therapeutics. Future work should further optimize brain-targeted delivery efficiency and action specificity of these agents, promote clinical translation, and explore their synergistic potential with mitophagy and mitochondrial-dynamics modulators to achieve more comprehensive therapeutic outcomes for AD.

4.2. Mitochondrial dynamics modulators

Marked mitochondrial dynamics imbalance is a prominent pathological feature in the early stage of AD, characterized by excessive mitochondrial fission and impaired fusion. Excessive fission induces mitochondrial fragmentation, a pathological change that not only reduces the efficiency of mitochondrial energy production but also disrupts the normal distribution of mitochondria at synaptic sites (Keskinoz et al., 2025). Therefore, inhibiting abnormal excessive mitochondrial fission and promoting its normal fusion has become a key intervention strategy for restoring the homeostasis of the mitochondrial network.

Among representative modulators in this field, Mdivi-1 is the most extensively studied mitochondrial fission inhibitor and a small-molecule inhibitor targeting DRP1. DRP1 is hyperactivated and translocates to the mitochondrial membrane in the brains of AD patients, thereby inducing abnormal mitochondrial fission (Manczak et al., 2019). Studies confirmed that Mdivi-1 treatment in AD mouse models can effectively inhibit the GTPase activity of DRP1, reduce mitochondrial fragmentation and restore the normal mitochondrial morphology. Meanwhile, it significantly decreases cerebral A β deposition and synaptic loss, and improves the learning and memory abilities of mice (Reddy et al., 2017, 2018). Another DRP1-targeted modulator is P110, a selective peptide inhibitor that specifically blocks the interaction between DRP1 and the mitochondrial receptor protein Fis1. Compared with Mdivi-1, P110 has higher targeting specificity and does not interfere with other physiological fission processes involving DRP1 (Qi et al., 2013). A study reported in EMBO Molecular Medicine that P110 treatment markedly reduces excessive mitochondrial fission and ROS production in AD mice, and effectively reverses synaptic dysfunction (Srivastava et al., 2023). In addition, the dynamin inhibitor Dynasore can also regulate the mitochondrial fission process by interfering with the GTPase activity of DRP1. Though its specificity is lower than that of P110, Dynasore has been proven to effectively prevent A β oligomer-induced mitochondrial fragmentation and protect the synaptic structure of neurons in in vitro neuronal models (Chen et al., 2019). As for modulators targeting the promotion of mitochondrial fusion, leflunomide is a highly representative candidate for drug repurposing. Through artificial intelligence (AI)-based drug screening technology, researchers found that this traditional anti-rheumatic drug can activate the mitochondrial fusion protein MFN2, whose downregulated expression in the brains of AD patients is an important cause of impaired mitochondrial fusion. Leflunomide improves the connectivity of the mitochondrial network by promoting MFN2-mediated outer mitochondrial membrane fusion, providing a novel approach for mitochondria-targeted therapy of AD

(Wang et al., 2021). S-Equol, a metabolite of soy isoflavones, also exhibits a dual regulatory effect on mitochondrial dynamics. Studies have found that it can significantly increase the expression of the mitochondrial fusion proteins Opa1 and Mfn1/2 while downregulating the expression level of DRP1. Through bidirectional regulation, it promotes mitochondrial elongation, enhances mitochondrial respiratory function, and effectively protects neurons in AD models from A β -induced toxic damage (Johnson et al., 2020; Yao et al., 2013). Besides the aforementioned synthetic or natural metabolite-based modulators, melatonin, an endogenous substance, acts as a potent mitochondrial protectant in addition to its classic role in sleep regulation. It can inhibit the phosphorylation of DRP1 at the Ser616 site - a key step in promoting mitochondrial fission - by regulating the calmodulin-dependent protein kinase II (CaMKII) signaling pathway, thereby blocking abnormal mitochondrial fission (Zhou et al., 2017). Preclinical studies have confirmed that long-term melatonin intervention can maintain the normal mitochondrial morphology and improve cognitive function in AD mice (Ebrahimi et al., 2025; Roy et al., 2022; Steinbach and Denburg, 2024). Furthermore, 17 β -estradiol regulates mitochondrial dynamics by binding to estrogen receptors (ERs), specifically upregulating the expression levels of MFN2 and OPA1 to promote mitochondrial fusion. This mechanism also provides a reasonable explanation for the increased incidence of AD in postmenopausal women and suggests that estrogen analogs are promising potential candidate drugs for regulating mitochondrial dynamics (Zhao et al., 2021).

In summary, mitochondrial dynamics imbalance represents a core early pathological hallmark of AD and a promising therapeutic target. Multiple modulators exert robust mitochondrial-protective and neuro-protective effects in pre-clinical models: targeting DRP1 to restrain excessive fission, activating MFN2/OPA1 to boost fusion, or achieving bidirectional regulation of mitochondrial dynamics (Table 3). These findings advance our mechanistic understanding of AD and offer multiple avenues for developing early-intervention agents. Future work should improve drug-targeting specificity, optimize administration regimens, and investigate combinatorial benefits with mitophagy modulators to foster breakthroughs in early-stage AD treatment.

4.3. Mitophagy enhancers

Mitophagy, a key quality control mechanism by which cells eliminate damaged and senescent mitochondria, exerts a crucial regulatory role in the pathological progression of AD (Skawratananond et al., 2025). Impaired function of the Pink1/Parkin pathway in the brains of AD patients leads to the abnormal accumulation of damaged mitochondria in neurons, which not only continuously release ROS and pro-apoptotic factors to exacerbate cellular damage but also serve as seeds for the aggregation of A β peptides and tau protein, further driving pathological progression (Tran and Reddy, 2020). Therefore, inducing mitophagy through drugs to accelerate the clearance of such abnormal mitochondria has become an important intervention direction for reversing AD pathological changes.

Among the research drugs in this field, urolithin A is a natural

Table 2
Mitochondria-targeted therapeutic drugs for AD – Mitochondrial mtDNA repair and protectants.

Representative drugs	Clinical status	Target / Core mechanism	Main pharmacological effects	References
NR/NMN	Phase I/II clinical trial	Activates SIRT1-PGC1 α -TFAM axis, elevates NAD ⁺ level.	Protects mtDNA, reduces oxidative damage, improves mitochondrial energy metabolism.	(Hou et al., 2018)
Vorinostat, Valproic acid	Phase I clinical trial; Phase II clinical trial	Upregulates OGG1/APE1, enhances mtDNA base excision repair (BER).	Reduces mtDNA oxidative damage.	(Ateya et al., 2025)
Latrepirdine (Dimebon)	Clinical development terminated	Blocks mPTP opening, protects mitochondrial membrane structure.	Reduces mtDNA mutation accumulation, resists A β toxicity.	(Zhang et al., 2010)
Berberine	Preclinical only	Activates AMPK-SIRT1-PGC1 α pathway.	Increases mtDNA copy number, elevates ATP level in hippocampal neurons.	(Akbar et al., 2021)
ApoE-mimetic peptide (COG1410)	Preclinical	Competitively inhibits ApoE4-mitochondria binding.	Alleviates mtDNA damage and cytotoxicity.	(Qiao et al., 2024)

Table 3
Mitochondria-targeted therapeutic drugs for AD – Mitochondrial dynamics modulators.

Representative drugs	Clinical status	Target / Core mechanism	Main pharmacological effects	References
Mdivi-1	Preclinical	Inhibits DRP1 GTPase activity, reduces excessive mitochondrial fission.	Attenuates mitochondrial fragmentation, decreases Aβ deposition and synaptic loss.	(Reddy et al., 2017)
P110	Preclinical	Selectively blocks DRP1-Fis1 interaction.	Inhibits abnormal fission, reduces ROS, reverses synaptic dysfunction.	(Srivastava et al., 2023)
Leflunomide	Approved (Market-launched)	Activates MFN2, promotes mitochondrial fusion.	Improves mitochondrial network structure and connectivity.	(Wang et al., 2021)
S-Equol	Phase II clinical trial	Upregulates fusion proteins (OPA1/MFN1/2), downregulates DRP1.	Promotes mitochondrial elongation, enhances respiratory function.	(Johnson et al., 2020)
Melatonin	Approved in some regions / Dietary supplement	Inhibits DRP1 Ser616 phosphorylation.	Inhibits DRP1 Ser616 phosphorylation, blocks abnormal mitochondrial fission, maintains morphological integrity.	(Zhou et al., 2017)

compound with great clinical translation potential. It is a metabolite of ellagitannins (found in pomegranates, berries and other foods) produced by intestinal microbiota. A study confirmed that urolithin A can potentially induce mitophagy by stabilizing Pink1 and Parkin proteins (Kshirsagar et al., 2025). In AD mouse models, this drug treatment significantly reduced the formation of cerebral Aβ plaques and hyperphosphorylation of tau protein, alleviated neuroinflammatory responses, and effectively improved the memory function of mice, making it one of the natural mitophagy inducers with the best clinical translation prospects at present (Kshirsagar et al., 2022; Pradeepkiran et al., 2022). Another drug with mitophagy-activating effects is the antibiotic actinonin, which is found to potentially activate mitophagy by mimicking the stress of the mitochondrial unfolded protein response (UPRmt) (Kshirsagar et al., 2021). In AD mouse models, actinonin can promote the lysosomal degradation of damaged mitochondria, restore axonal mitochondrial transport function, and improve synaptic morphological integrity (Pradeepkiran et al., 2022). Spermidine, a natural polyamine with remarkable anti-aging properties, has also become a research hotspot in this field. It induces the expression of autophagy-related (ATG) genes by inhibiting the acetyltransferase EP300, thereby promoting global autophagy and mitophagy (Pietrocola et al., 2015). Studies on AD models have shown that spermidine can cross the blood-brain barrier, clear soluble Aβ, and enhance mitochondrial respiratory function (Freitag et al., 2022; Youssef et al., 2024). Rapamycin, a classic mTOR inhibitor, functions by regulating the initiation of autophagy: it relieves the inhibitory effect of mTOR on the autophagy initiation complex ULK1, thus activating autophagy (Hosokawa et al., 2009). Multiple studies have confirmed that it can reduce Aβ and tau pathological changes in the brains of AD mice (Caccamo et al., 2010; Spilman et al., 2010), but its long-term clinical application is limited due to immunosuppressive side effects. The current research and development direction focuses on developing brain-specific or better-tolerated rapalogs (rapamycin analogs) (Eshraghi et al., 2022). Metformin, a first-line antidiabetic drug, induces mitophagy by activating the AMP-activated protein kinase (AMPK) pathway. AMPK can not only directly phosphorylate ULK1 to initiate autophagy but also phosphorylate outer mitochondrial membrane receptors to promote the recognition of mitochondria by autophagosomes (Kim et al., 2011). Studies have also suggested that diabetic patients taking metformin for a long time have a significantly reduced risk of developing AD (Khezri et al., 2022; Luo et al., 2022). Resveratrol, a natural polyphenol, promotes both Pink1/Parkin-mediated mitophagy and receptor-mediated mitophagy (e.g., the BNIP3L/Nix pathway) by activating the SIRT1 signal, which can accelerate the turnover of damaged mitochondria and reduce neuronal death (Das et al., 2014). Kaempferol, a flavonoid compound, also exhibits mitophagy-enhancing activity. A study has confirmed that it can enhance mitophagic flux, protect SH-SY5Y cells from Aβ-induced toxic damage, and delay Aβ-induced paralysis phenotype in *Caenorhabditis elegans* models (Xie et al., 2022). It is worth noting that the development of mitophagy enhancers needs to focus on two core issues: first, avoiding autophagic cell death caused by “excessive autophagy”; second, ensuring lysosomal

functional integrity - efficient fusion of autophagosomes with lysosomes is essential for the degradation of damaged mitochondria. Natural compounds such as urolithin A have become the current research hotspots and key directions for clinical translation in this field due to their superior safety profile (Hou et al., 2024).

To conclude, various mitophagy modulators target and improve AD-related pathological damage through different signaling pathways (Pink1/Parkin, AMPK, SIRT1, etc.), providing diverse strategies for AD treatment (Table 4). Among them, natural compounds have the advantage of safety, classic drugs are backed by clinical application foundations, and plant active ingredients show multi-target synergistic potential. These studies provide important experimental evidence for the clinical translation of AD therapies. Future research needs to further focus on the brain targeting, side effect optimization and clinical efficacy verification of drugs to drive breakthrough progress in this field.

4.4. Mitochondrial transplantation technology

When the endogenous mitochondrial repair mechanism malfunctions, mitochondrial replacement/transplantation technology, which involves exogenous supplementation of healthy mitochondria, has emerged as a transformative intervention strategy in the field of AD treatment (Chang et al., 2019). The core principle of this technology is to introduce functional mitochondria isolated from healthy cells such as mesenchymal stem cells (MSCs) into damaged neurons, and reverse the pathological state of AD by directly reconstructing the cellular energy supply system and restoring neuronal viability (Nitzan et al., 2019). At present, the main technical approaches and delivery vectors explored in this field are developing in a diversified manner, and all strategies focus on improving the delivery efficiency, targeting ability and functional stability of exogenous mitochondria.

Studies have shown that direct intracerebral stereotaxic injection is the most basic delivery method. Researchers inject isolated mitochondria extracted from healthy tissues directly into the hippocampal or cortical regions of AD mice. It has been found that neurons can take up these exogenous mitochondria through endocytosis, and the exogenous mitochondria can survive and maintain normal functions in recipient cells. This approach not only significantly increases intracellular ATP levels and reduces oxidative stress damage, but also effectively improves the cognitive behaviors of model animals in the short term (Bobkova et al., 2022; Zhang et al., 2022). To further enhance the efficiency of exogenous mitochondria in penetrating cell membranes, researchers have developed a mitochondria delivery strategy modified with cell-penetrating peptides, with PEP-1 peptide-modified mitochondria being a representative technology in this direction. Surface modification of isolated healthy mitochondria with PEP-1 peptide enables mitochondria to enter AD model cells and brain tissues more efficiently, achieve functional replacement of damaged mitochondria, effectively restore the activity of the electron transport chain, and reverse Aβ-induced neurotoxic damage (Chang et al., 2017, 2013). The delivery strategy based on stem cell-derived extracellular vesicles

Table 4
Mitochondria-targeted therapeutic drugs for AD – Mitophagy enhancers.

Representative drugs	Clinical status	Target / Core mechanism	Main pharmacological effects	References
Urolithin A	Phase II clinical trial	Stabilizes PINK1/Parkin, strongly induces mitophagy.	Clears damaged mitochondria, reduces A β and tau pathology.	(Kshirsagar et al., 2022)
Actinonin	Preclinical	Activates mitochondrial unfolded protein response (UPRmt).	Promotes degradation of damaged mitochondria, restores axonal transport.	(Kshirsagar et al., 2021)
Spermidine	Phase II clinical trial	Inhibits EP300, induces autophagy/mitophagy.	Clears soluble A β , enhances mitochondrial respiratory function.	(Freitag et al., 2022)
Rapamycin	Approved (Immunosuppression) / Phase II/III for anti-aging	Inhibits mTOR, activates autophagy pathway.	Reduces A β and tau deposition in the brain.	(Caccamo et al., 2010)
Metformin	Approved (Diabetes) / Phase III for anti-aging	Activates AMPK pathway, induces mitophagy.	Reduces AD risk in diabetic patients.	(Khezri et al., 2022)
Resveratrol	Phase II clinical trial	Activates SIRT1, promotes PINK1/Parkin-mediated mitophagy.	Accelerates turnover of damaged mitochondria, reduces neuronal death.	(Das et al., 2014)

(EVs)/exosomes relying on the cellular paracrine mechanism has become an important research direction for non-invasive mitochondrial transplantation. MSCs can release EVs containing functional mitochondria or mtDNA through paracrine pathways, and these vesicles can cross the BBB and be taken up by neurons smoothly. Current related research is focusing on loading healthy mitochondria into engineered exosomes to achieve efficient and non-invasive delivery of exogenous mitochondria by virtue of this natural nanocarrier (Ghosh and Pearse, 2024; Phinney et al., 2015). MitoCeption technology provides an exclusive method for in vitro mitochondrial transfer, which uses centrifugal force combined with co-incubation to transfer healthy mitochondria into target cells. Although this technology is currently mainly applied in cell experiments, it has laid an important technical foundation for subsequent strategies such as modifying autologous cells including induced pluripotent stem cell (iPSC)-derived neural progenitor cells and re-infusing them for treatment after mitochondrial transplantation (Sercel et al., 2021). Mitochondrial transfer mediated by intercellular tunneling nanotubes (TNTs) is a novel strategy based on the physiological mechanism of intercellular material exchange. Studies have found that when healthy MSCs are co-cultured with damaged neurons, MSCs can spontaneously form TNTs and directionally transport healthy mitochondria to damaged neurons. Current research strategies in this field include screening small-molecule drugs such as Rho kinase inhibitors to enhance the mitochondrial support of endogenous or transplanted cells to damaged neurons by promoting the formation of TNTs (Zurzolo, 2021). In addition, aiming at the core pathological problem of energy deficiency at synaptic sites in the early stage of AD, researchers have also developed a synaptosome-targeted delivery strategy, which uses synaptosomes as specific carriers to accurately deliver healthy mitochondria to synaptic sites, improve the functional defects of synaptic mitochondria in a targeted manner, and alleviate synaptic energy metabolic disorders (Picone et al., 2021; Toader et al., 2024).

Mitochondrial transplantation represents a promising mitochondria-targeted therapeutic candidate for AD. Preclinical data support its capacity to replenish functional mitochondria, restore energy metabolism and suppress tau hyperphosphorylation and aggregation. Nevertheless, this field suffers major methodological concerns (Hole et al., 2026). Devine's group has uncovered critical pitfalls in common mitochondrial-transfer assays. Widely used lipophilic cationic dyes (e.g., Mitotracker Red/Green) suffer from non-specific diffusion, medium retention, failure to discriminate intact mitochondria from fragments, and non-linear fluorescence-quantity correlation. Consequently, co-culture systems produce ~80% false-positive signals, leading to a 2- to 5-fold overestimation of transfer efficiency. Studies adopting Mitotracker labeling — including dye-tagged mitochondrial direct injection and PEP-1-modified mitochondria assessed via dyes — generate uncertain estimates for transfer efficiency and therapeutic effects and demand stricter validation. By comparison, genetic labeling approaches, such as mtDNA haplotype sequencing, mt-Keima and mito-GFP, bypass dye-associated artifacts and reliably verify the transfer and functional

integration of intact mitochondria within recipient cells.

At present, mitochondrial transplantation technology is still in the preclinical research stage as a whole. Its clinical translation faces core challenges such as the immunogenicity of exogenous mitochondria, the delivery efficiency across the blood-brain barrier, and the long-term survival of exogenous mitochondria in recipient cells. However, adhering to the therapeutic logic of directly compensating for mitochondrial functional defects, this technology provides a brand-new therapeutic hope for advanced AD patients with severe mitochondrial functional failure.

4.5. Clinical challenges of mitochondria-targeted drugs

Mitochondria-targeted interventions yield promising neuroprotective effects in AD preclinical models, yet most candidates entering clinical trials miss primary endpoints, illustrating the translational gap between bench and clinic. A representative example is latrepirdine (Dimebon), the first mitochondrial-protective drug reaching Phase III. Though it improved cognition in Phase II, two large Phase III trials showed no benefit versus placebo. Post-hoc analyses link its failure to ill-defined targets, delayed intervention, and high patient heterogeneity (Bezprozvanny, 2010; Jones, 2010). MitoQ, the most representative mitochondria-targeted antioxidant, has exhibited excellent neuroprotective effects in multiple AD animal models, yet in Phase II clinical trials enrolling patients with mild to moderate AD, it failed to significantly improve cognitive function and activities of daily living; the main reasons for its failure likely include insufficient brain exposure of the drug, inability to block the complex pathological network by targeting only the single pathway of oxidative stress, and irreversible neuronal damage already present in enrolled participants (Persson et al., 2014). Although SS-31 (Elamipretide) has demonstrated potent mitochondrial protective effects in various neurodegenerative disease models, its Phase II clinical trial for AD also failed to meet the primary cognitive endpoint; analyses suggest its failure may be related to poor patient compliance due to the intravenous administration route, inadequate treatment duration, and the absence of patient stratification based on mitochondrial biomarkers (Mi et al., 2021). These clinical setbacks do not invalidate the core rationale for mitochondria-targeted therapy; instead, they highlight four key challenges in current research. First, non-invasive biomarkers reflecting cerebral mitochondrial function are lacking, hindering accurate screening of treatment-responsive patients. Second, interventions are frequently administered too late: most trials initiate treatment after overt cognitive impairment emerges, when substantial irreversible neuronal and synaptic loss has already taken place. Third, most agents target only one node within the mitochondrial functional network and cannot disrupt the vicious pathological cycle driven by intertwined AD mechanisms such as oxidative stress, energy metabolic dysfunction and neuroinflammation. Fourth, insufficient recognition of AD pathological heterogeneity leads to a “one-size-fits-all” therapeutic paradigm, which overlooks inter-patient differences

in predominant mitochondrial-dysfunction mechanisms.

5. Limitations of current research

Although remarkable progress has been made in recent years in research on the association between mitochondrial dysfunction and AD, and the notion that mitochondrial senescence acts as a key upstream driver of the disease has gained widespread recognition, current research still faces numerous profound limitations. These limitations have greatly hindered the in-depth elucidation of disease mechanisms and the efficient translation of basic research findings into clinical applications. One of the core challenges is the overreliance on disease models, as well as the high heterogeneity between these models and the pathology of human SAD.

At present, the vast majority of mechanistic studies and drug screening efforts rely heavily on transgenic animal models carrying FAD gene mutations, such as APP/PS1 or 3xTg-AD mice. While such models can effectively recapitulate classic pathological hallmarks including A β deposition and tau hyperphosphorylation, the temporal onset of mitochondrial dysfunction in these models differs fundamentally from that in human AD (Yao et al., 2009). A wealth of clinical evidence has shown that in human SAD, signs of reduced cerebral glucose metabolism and mitochondrial functional decline may emerge decades prior to the widespread appearance of A β plaques, which directly supports the core hypothesis that mitochondrial dysfunction is the initiating factor of AD (Cha et al., 2026; Di Domenico et al., 2024; Motamed et al., 2026; Prabha et al., 2025). In transgenic mouse models, however, mitochondrial dysfunction typically manifests significantly only after the peak of A β pathology, representing a secondary alteration. This may lead to reversed causal inferences in research conclusions derived from such models, which regard mitochondrial damage merely as a downstream consequence of A β toxicity, thereby underestimating its true value as an upstream therapeutic target (Hauptmann et al., 2009). More crucially, interspecies biological differences directly compromise the generalizability of research findings. Mice and humans exhibit notable disparities in mitochondrial genome composition, the subunit structure of OXPHOS complexes, as well as the sequence and expression patterns of mitochondrial dynamics regulatory proteins (e.g., DRP1, MFN2) (Wallace, 2005). For instance, APOE ϵ 4, the strongest genetic risk factor for SAD, induces mitochondrial dysfunction in human neurons that cannot be fully recapitulated in standard mouse models (Chen et al., 2025). Such discrepancies directly result in a disconnect between the performance of drugs in preclinical and clinical studies: a typical example is the mitochondrial fission inhibitor Mdivi-1, which exerts significant neuroprotective effects in APP/PS1 mice but shows markedly reduced inhibitory efficiency and obvious cytotoxicity in human neuronal cell lines (Bordt et al., 2017). This prompts us to interpret research findings based on animal models with extreme caution.

In addition, although induced pluripotent stem cell (iPSC) technology has provided an unprecedented window for human SAD research, it has its own inherent limitations. Neurons reprogrammed from somatic cells of elderly patients have their "aging clock" partially reset, which may prevent them from fully recapitulating the mitochondrial state of genuine aged neurons that have undergone decades of cumulative damage (Gudenschwager et al., 2021; Sichani et al., 2024). Accordingly, iPSC models have critical limitations in recapitulating the long-term chronic course of mitochondrial senescence. For clinical translation, major gaps remain in biomarker development, targeted-delivery system construction, and consideration of disease heterogeneity. Non-invasive, precise biomarkers (e.g., novel PET tracers) reflecting in-vivo human-brain mitochondrial function are urgently required for early high-risk-population screening and objective therapeutic-response assessment. Nevertheless, the direct correlation of existing peripheral-blood- or cerebrospinal-fluid-derived biomarkers with cerebral pathological changes still requires further validation (Amanullah et al., 2025). Meanwhile, the development of drug delivery systems

that can efficiently cross the blood-brain barrier and precisely target neuronal mitochondria is a critical bottleneck for the successful clinical translation of mitochondria-targeted drugs (Zhang et al., 2023). More importantly, SAD itself is a highly heterogeneous syndrome, and mitochondrial dysfunction in different patients may be driven by distinct dominant mechanisms: some cases are mainly characterized by a high mtDNA mutation load, while others center on mitochondrial dynamics imbalance or autophagic defects (Modiri et al., 2025). The current one-size-fits-all therapeutic strategies ignore such individual differences, which may be one of the important reasons for the failure of numerous clinical trials for AD.

Therefore, future research efforts should focus on developing novel disease models that more closely mimic the pathology of human SAD, establishing a reliable system of in vivo mitochondrial biomarkers, and implementing precise, multi-target combined intervention strategies based on patients' pathological subtypes. Only in this way can the promising vision of mitochondria-targeted therapy be truly translated into effective treatments that can slow down or even reverse the progression of AD.

6. Future directions and prospects

Given the complex pathological mechanisms of mitochondrial dysfunction in AD, traditional intervention strategies are no longer sufficient to meet clinical needs, and there is an urgent need to explore innovative and targeted novel approaches. With the in-depth integration of life sciences and cutting-edge technologies, intervention strategies targeting mitochondrial senescence have gradually emerged, providing a crucial opportunity to break through the bottleneck of AD treatment. Among these, cell reprogramming technologies have become a highly researched hotspot due to their unique senescence-reversing properties.

Addressing this core pathological cycle, cell reprogramming technologies (e.g., iPSC reprogramming, partial reprogramming) offer a completely new direction for AD intervention (Odonchimed et al., 2026). Their core advantage lies in the ability to directly reset the cellular aging clock and reverse mitochondrial senescence-associated dysfunction from the root, rather than achieving partial repair of a single target. Specifically, iPSC reprogramming can reprogram somatic cells (e.g., skin fibroblasts) from AD patients into pluripotent stem cells, which are then further differentiated into neurons (Israel et al., 2012). Studies have confirmed that neurons obtained through such reprogramming exhibit a significant reversal of inherent mitochondrial senescence features (e.g., elevated mtDNA mutation load, OXPHOS dysfunction), laying the foundation for replacing damaged neurons in the AD brain and reconstructing normal mitochondrial function. Of greater clinical translation potential is partial reprogramming technology: this approach briefly activates the Yamanaka reprogramming factors (Oct4, Sox2, Klf4, c-Myc) without inducing complete cellular dedifferentiation, enabling the reversal of aging-associated phenotypes while preserving the original cellular identity. Research has verified that this method can effectively improve mitochondrial function, restore energy metabolism efficiency and extend lifespan in progeroid mice (Ocampo et al., 2016; Suhr et al., 2010). In the field of AD, partial reprogramming is believed to hold the potential to systematically reverse defects in mitochondrial biogenesis and reduce mtDNA mutation load in senescent neurons by resetting the cellular epigenetic clock (e.g., restoring a youthful DNA methylation pattern). Its repair efficacy is likely to surpass single metabolite supplementation or single-target drugs (Sarkar et al., 2020).

In addition to cell reprogramming technologies, the rise of AI has provided a revolutionary tool for mitochondria-targeted drug discovery and development, effectively solving the pain points of long development cycles and low efficiency in traditional drug design, and emerging as another important direction for AD intervention (Xie et al., 2022). AI algorithms can deeply mine massive biological datasets and accurately predict drug targets that are difficult to identify via traditional methods,

such as allosteric sites at protein-protein interaction interfaces. In particular, deep learning tools such as AlphaFold2 can predict the 3D structure of proteins with unprecedented precision, enabling the identification of cryptic allosteric pockets on mitochondrial dynamics proteins (e.g., Drp1, Mfn2) and thus breaking the bottleneck of drug design for such flat PPI targets (Malhotra et al., 2025). While AI-driven drug design represents the future direction, advanced screening technologies have already validated the feasibility of targeting mitochondrial dynamics proteins. A groundbreaking study using high-throughput screening successfully identified a novel small-molecule agonist of Mfn2. This molecule can specifically activate the GTPase activity of Mfn2, significantly restore mitochondrial fusion, improve synaptic plasticity, and ultimately rescue cognitive deficits in AD mouse models (Li, 2025). This successful case paves the way for the future development of more potent analogs designed with AI. This successful case paves the way for the future development of more potent analogs designed with AI. Furthermore, machine learning algorithms analyzing peripheral blood mitochondrial gene expression profiles can accurately stratify AD patients into three molecular subtypes: oxidative stress subtype, energy metabolism subtype, and autophagy-deficient subtype, which exhibit significant heterogeneity in therapeutic responses to mitochondria-targeted drugs (Ma et al., 2023). For instance, AD patients identified with a predominant “mitophagy-deficient” subtype could be preferentially assigned to clinical trials for mitophagy enhancers (e.g., Urolithin A or NAD+ precursors), whereas those exhibiting an “oxidative stress-dominant” profile would benefit more from mitochondria-targeted antioxidants (e.g., MitoQ). This precision medicine paradigm ensures that mitochondria-targeted agents are administered to the most biologically responsive patient subgroups at the optimal therapeutic window, thereby maximizing clinical efficacy and circumventing the failures caused by population heterogeneity. On this basis, AI models can predict the efficacy and safety of specific mitochondria-targeted drugs in individual patients, providing a basis for precise subject selection in clinical trials and is expected to significantly improve the success rate of AD clinical trials (More et al., 2026). In the realm of biomarker discovery, AI frameworks integrating multi-omics data have achieved remarkable success. For instance, researchers utilized machine learning algorithms to analyze blood metabolite profiles from a large cohort, successfully identifying a robust panel of metabolic biomarkers that can diagnose AD with high accuracy. This AI-driven approach directly bridges systemic metabolic and mitochondrial alterations with central pathology (Stamate et al., 2019). Similarly, deep learning models applied to longitudinal clinical and neuroimaging data have proven highly effective in tracking and predicting the preclinical progression of AD, offering a critical temporal window for early mitochondria-targeted interventions (Lee et al., 2019). In terms of treatment regimen prediction and drug discovery, AI provides a powerful tool to overcome the high clinical heterogeneity of AD. A

landmark example is the AI-based network medicine framework developed by researchers, which integrated large-scale genomic data and human interactomes to predict drug efficacy. By analyzing electronic health records from over 7 million patients, their AI model successfully predicted and validated that sildenafil—a drug modulating energy metabolism and vascular function—could significantly reduce the risk of AD (Fang et al., 2021).

Traditional interventions for AD have shown limited efficacy (Table 5), while cell reprogramming technologies can reset the cellular aging clock to reverse mitochondrial dysfunction—with partial reprogramming boasting greater clinical potential. AI technologies break through the bottlenecks of traditional drug research and development by accurately mining drug targets and designing targeted agents. The synergy of these two approaches provides an innovative and highly efficient new direction for AD treatment.

7. Conclusion

This review summarizes the link between mitochondrial dysfunction and AD pathogenesis and identifies mitochondrial senescence as a critical upstream node for AD initiation. Its hallmarks-impaired biogenesis, defective quality control, and disrupted functional homeostasis-jointly promote AD progression via multiple mechanisms: impaired oxidative phosphorylation, mtDNA mutations, excessive ROS generation, mitochondrial dynamics imbalance, mitophagy dysfunction, calcium homeostasis disturbance, and mitochondrial transport defects. Rather than acting independently, these mechanisms build a complex regulatory network through cascade amplification, mutual modulation, and bidirectional crosstalk with the aged AD microenvironment, underpinning AD pathological advancement. Given the central role of mitochondrial dysfunction, mitochondria-targeted interventions exhibit promising potency in preclinical and clinical studies and offer novel insights for AD therapeutics.

Abbreviations

AD, Alzheimer’s disease; mtDNA, Mitochondrial DNA; OXPHOS, Oxidative phosphorylation; ROS, Reactive oxygen species; AI, Artificial intelligence; Aβ, β-amyloid; APP, Amyloid precursor protein; NFTs, Neurofibrillary tangles; MCU, Mitochondrial calcium uniporter; NCLX, Mitochondrial sodium-calcium exchanger; MAMs, Endoplasmic reticulum-mitochondrial contact sites; mPTP, Mitochondrial permeability transition pore; AIF, Apoptosis-inducing factor; IAPs, Inhibitor of apoptosis proteins; PINK1, PTEN-induced kinase 1; Parkin, Parkinson disease protein 2; LC3, Microtubule-associated protein 1 light chain 3; BNIP3, BCL2/adenovirus E1B 19 kDa protein-interacting protein 3; NIX, BNIP3-like; FUNDC1, FUN14 domain-containing protein 1; MFN1, Mitofusin 1; MFN2, Mitofusin 2; OPA1, Optic atrophy 1; DRP1,

Table 5
Comparison of preclinical and clinical progress in mitochondria-targeted therapeutic strategies for AD.

Therapeutic strategy	Representative agents/Tech	Preclinical efficacy	Clinical status	Translational bottlenecks
Mitochondria-targeted antioxidants	SS-31, Astaxanthin	Attenuate neuroinflammation, improve synaptic function.	Most advanced (Phase II/III)	Low brain-targeted delivery efficiency; poor long-term systemic bioavailability.
mtDNA repair & protectants	NAD+ precursors (NMN/NR)	Reduce mtDNA mutations, restore energy metabolism.	Early clinical (Phase I/II)	Lack of specific <i>in vivo</i> biomarkers reflecting brain mtDNA repair status.
Mitochondrial dynamics modulators	Mdivi-1 (Drp1 inhibitor)	Block mitochondrial fragmentation, rescue cognition.	Severely lagging (Preclinical only)	Drp1 is ubiquitous; systemic administration risks severe off-target toxicity.
Mitophagy enhancers	Urolithin A, Spermidine	Significantly reduce Aβ and tau pathologies.	Under validation (Phase II)	Narrow therapeutic window with risk of “excessive autophagy” induced cell death; highly dependent on lysosomal function.
Mitochondrial transplantation	MSC-derived mitochondria	Directly reconstruct cellular energy supply systems.	No AD trials (Proof-of-concept)	Difficulties in standardized extraction; extremely low brain delivery efficiency; immune rejection and ethical controversies.

Dynamin-related protein 1; FIS1, Fission 1; MFF, Mitochondrial fission factor; KIF5A, Kinesin family member 5A; KIF5B, Kinesin family member 5B; Miro1, Mitochondrial Rho GTPase 1; Milton, TRAK1/2-interacting protein; TRAK1/2, Trafficking kinesin-binding protein 1/2; SOD1, Superoxide dismutase 1; SOD2, Superoxide dismutase 2; BACE1, β -secretase 1; γ -secretase, Gamma-secretase; IDE, Insulin-degrading enzyme; NEP, Neprilysin; GSK3 β , Glycogen synthase kinase 3 β ; CDK5, Cyclin-dependent kinase 5; CaMKII, Calmodulin-dependent protein kinase II; NF- κ B, Nuclear factor kappa-light-chain-enhancer of activated B cells; TLR9, Toll-like receptor 9; TNF- α , Tumor necrosis factor-alpha; IL-1 β , Interleukin-1 β ; iNOS, Inducible nitric oxide synthase; NO, Nitric oxide; PGC-1 α , Peroxisome proliferator-activated receptor γ coactivator 1 α ; NRF1, Nuclear respiratory factor 1; PI3K-AKT, Phosphatidylinositol 3-kinase-protein kinase B; JNK, c-Jun N-terminal kinase; AMPK, AMP-activated protein kinase; SIRT1, Sirtuin 1; TFAM, Mitochondrial transcription factor A; BER, Base excision repair; OGG1, 8-oxoguanine DNA glycosylase 1; APE1, Apurinic/aprimidinic endonuclease 1; HDACi, Histone deacetylase inhibitors; ApoE4, Apolipoprotein E4; COG1410, ApoE-mimetic peptide; MTA, Mitochondria-targeted antioxidant; TPP⁺, Triphenylphosphonium; MitoQ, Mitochondrial coenzyme Q10; Mito-Tempo, Mitochondrial Tempo; MitoVitE, Mitochondrial Vitamin E; SkQ1, Plastoquinone-decylidiphenylphosphonium; SS-31, Szeto-Schiller peptide 31; BBB, Blood-brain barrier; LTP, Long-term potentiation; Erk1/2, Extracellular signal-regulated kinase 1/2; CREB, cAMP response element-binding protein; Mdivi-1, Mitochondrial division inhibitor 1; P110, DRP1-FIS1 interaction inhibitor peptide; Dynasore, Dynamin inhibitor; Leflunomide, Anti-rheumatic drug; S-Equol, Soy isoflavone metabolite; UPRmt, Mitochondrial unfolded protein response; ATG, Autophagy-related; EVs, Extracellular vesicles; iPSC, Induced pluripotent stem cell; MSC, Mesenchymal stem cell; PEP-1, Cell-penetrating peptide; TNTs, Tunneling nanotubes; Rho kinase, Rho-associated protein kinase; FAD, Familial Alzheimer's disease; SAD, Sporadic Alzheimer's disease; PET, Positron emission tomography; Oct4, Octamer-binding transcription factor 4; Sox2, SRY-box transcription factor 2; Klf4, Kruppel-like factor 4; c-Myc, Myelocytomatosis oncogene; NR, Nicotinamide riboside; NMN, Nicotinamide mononucleotide. IDE, Insulin-degrading enzyme; NEP, Neprilysin.

Author contributions

XL, RW and CL designed and wrote the manuscript. XL, RW and YW produced figures. YZ, XC, YL, RY, DZ and QL edited the manuscript. QL and CL designed and edited the manuscript. All authors have read and approved the final version of the manuscript.

Informed consent statement

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Declaration of Competing Interest

We declare that we have no financial and personal relationships with

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Data Availability

No data was used for the research described in the article.

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